



HOKKAIDO
UNIVERSITY

Propagation of Mind Through the Mechanism of Superposition

**Center for Human nature, AI, and Neuroscience (CHAIN),
Hokkaido University, Japan**



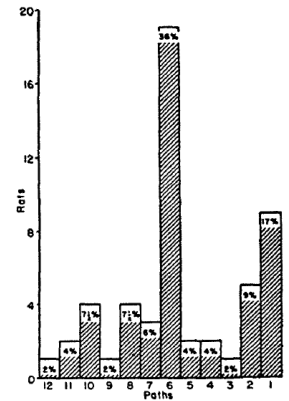
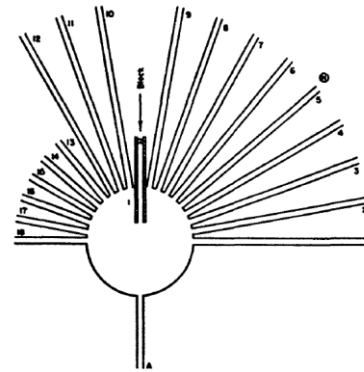
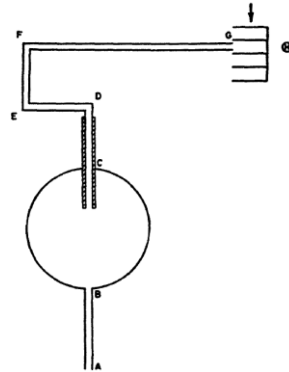
Center for Human Nature,
Artificial Intelligence, and Neuroscience

Hiroyuki Iizuka

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Spatial recognition and neural activation

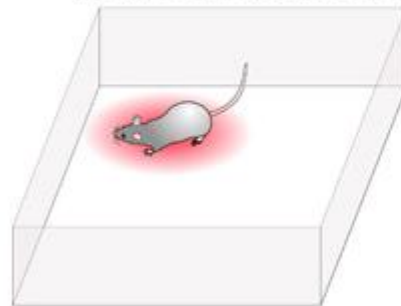
Behavior



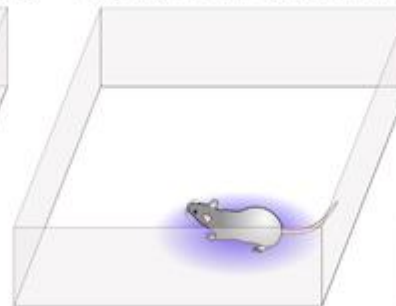
[Tolman, 1948]

Neural activation

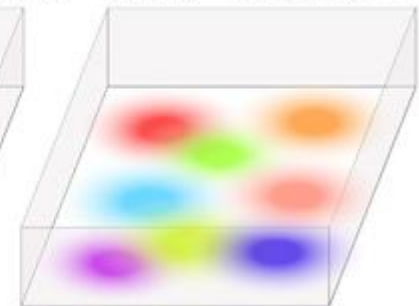
場所細胞 1 は、
ねずみがこの位置に来たときに活動



場所細胞 2 は、
ねずみがこの位置に来たときに活動



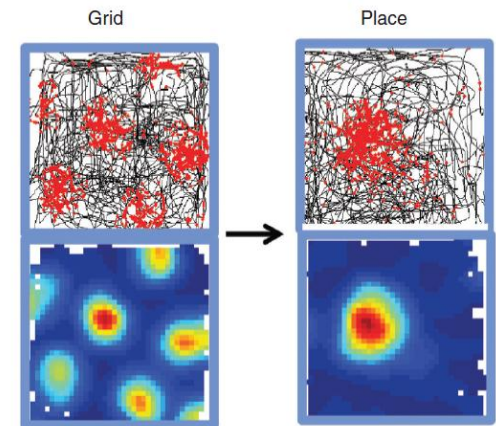
場所細胞がたくさん集まると
空間のすべての領域をカバー



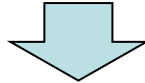
[日本神経科学会web]

Place Cell

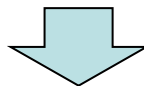
- **Place cell:**
neurons activated when animals enter in a particular place.
[O'Keefe & Dostrovsky, 1971]
- **Grid cell:**
neurons activated at an equal distances.
[Fyhn et al., 2004, Hafting et al. 2005]



[Moser et al. 2015]



- Mathematical models have been proposed that retain spatial representation.
- Rats and infants are never told where they are.
- No model where spatial representation is built from only visuo-motor experiences.

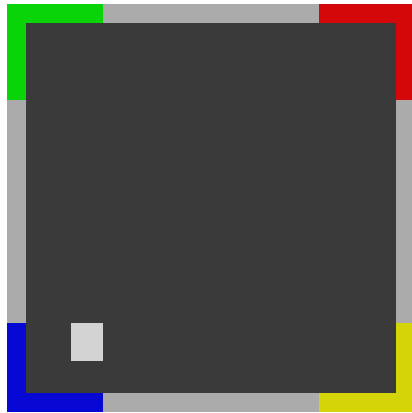


Simulation-based synthetic approach using deep neural networks

Predictive learning

- **Simulation environment**

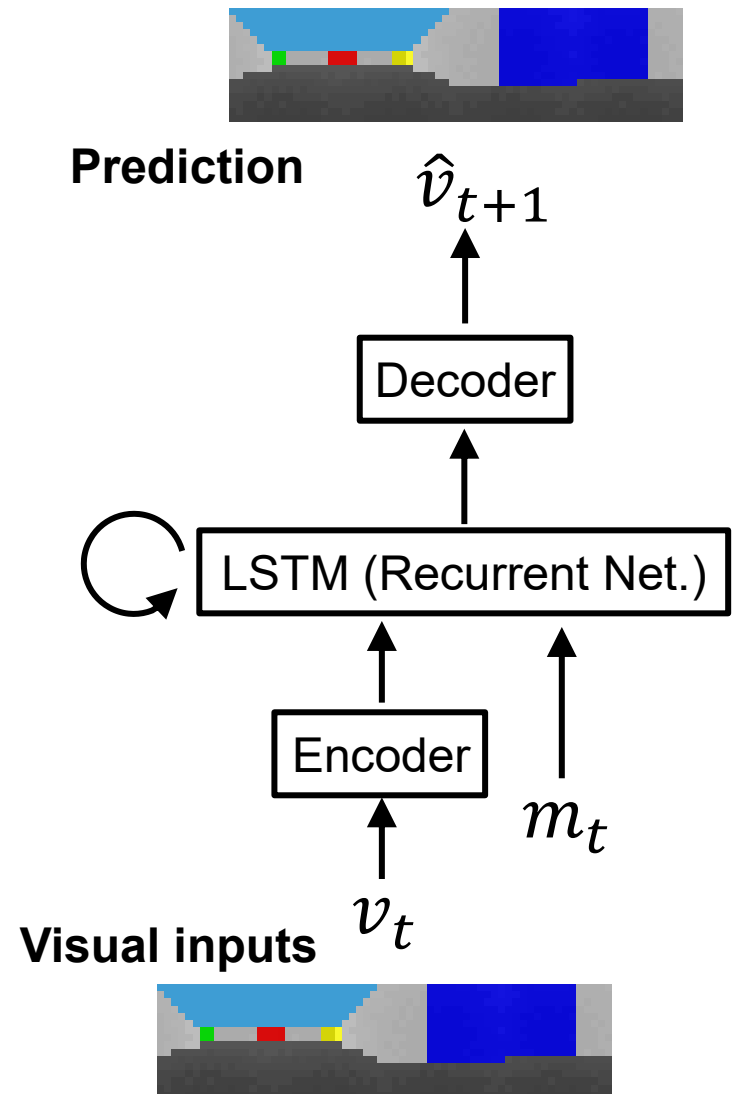
Agent : a robot moving around 2-D space



➤ **Four landmarks at the corners**

The neural network is trained to **predict** future visual inputs.

- **Neural network model**

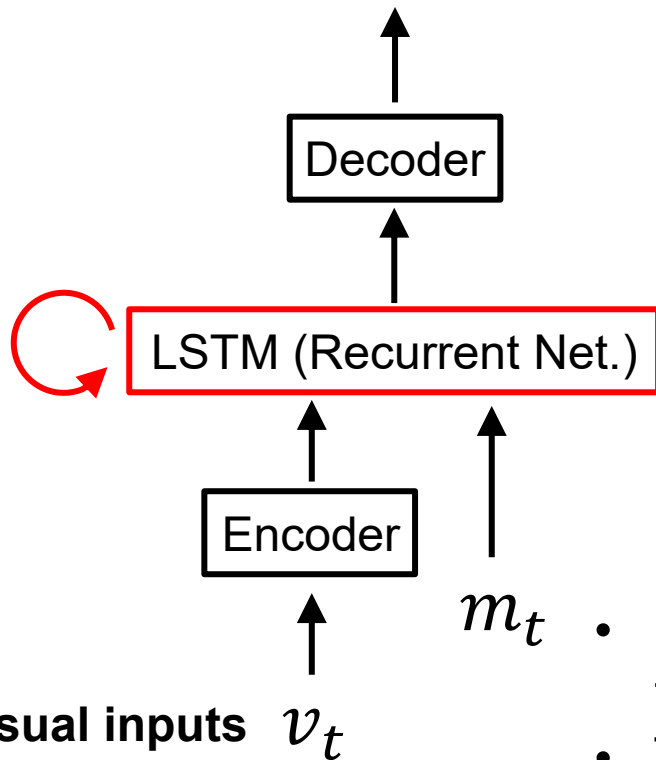


Visualizing internal states

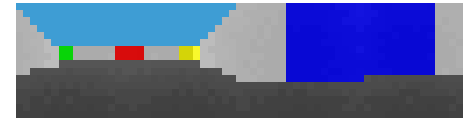
Prediction performance

Prediction

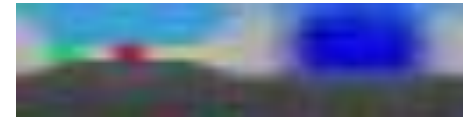
$\hat{v}_{t+1}$



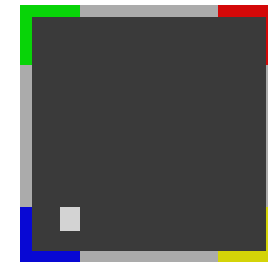
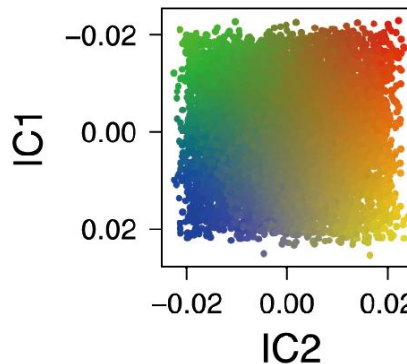
Ground Truth



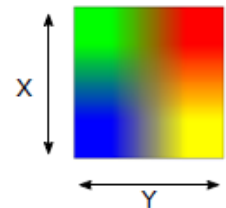
Prediction



Visualized internal states



Color map



- Internal states are stored in such a way that they correspond to the locations.
- Two dimensions enough same as spatial dimensions.

Real-world experiments

Prediction performance

Prediction

$\hat{v}_{t+1}$

Decoder

LSTM (Recurrent Net.)

Encoder

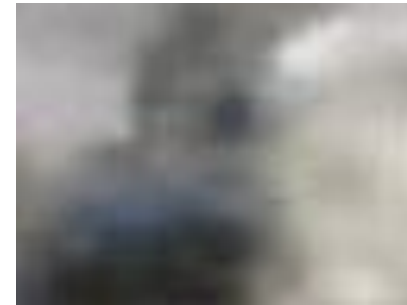
m_t

Visual inputs v_t

Ground Truth



Prediction



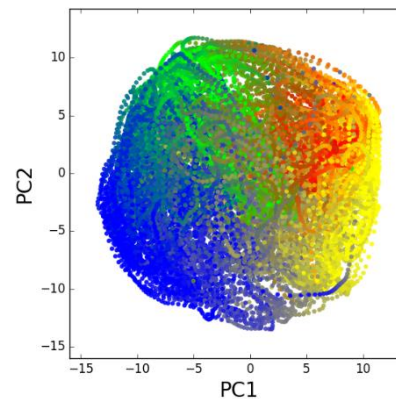
acc.



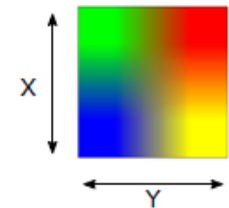
camera



Visualized internal states



Color map



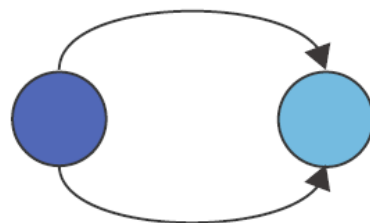
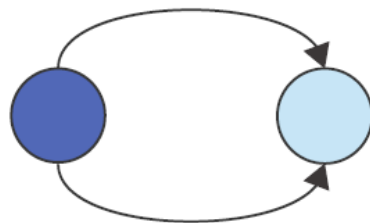
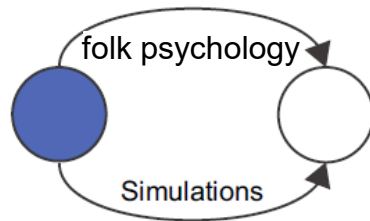
- **Distinction of self and others through superposition mechanism**
- **Propagating a mind from other to self via superposition mechanism**

How do we predict and explain other's behavior?

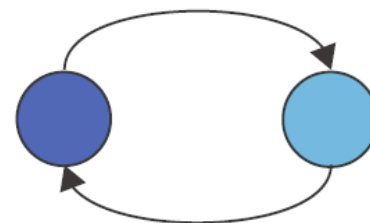
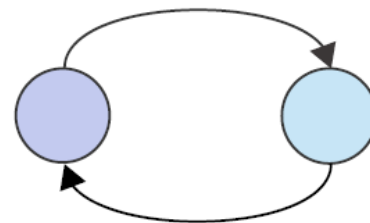
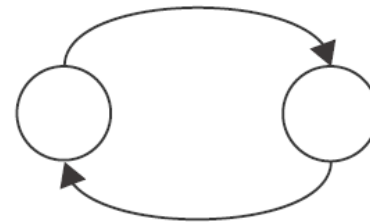
- Theory theory [Gopnik, 1994 ...]
 - Using implicit general knowledge called folk psychology
 - Knowledge-based inference using beliefs and desires
- Simulation theory [Gordon, 1986 ...]
 - Using its own model to simulate other's mental states.
 - Model-based inference
- Interaction theory [Gallagher, 2008 ...]
 - Criticizing the view of unilateral observation of others and making inferences about others' mental states
 - Ongoing interaction itself causes direct perception of others

Illustration of those approaches

Theory theory Simulation approach



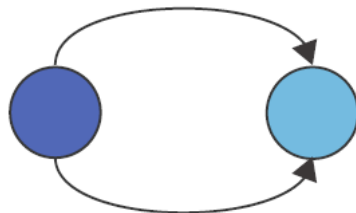
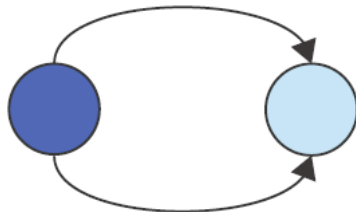
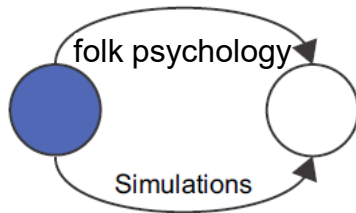
Interaction theory



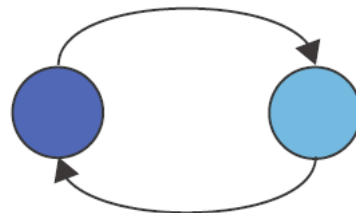
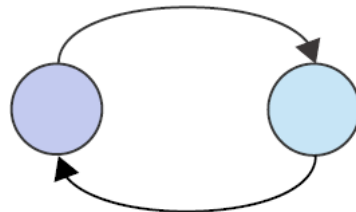
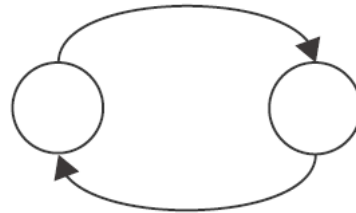
How can the pre-existing frameworks of self and others be formed?

No pre-existing frames of self and other

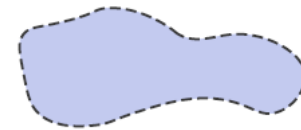
TT and ST



IT



Our idea

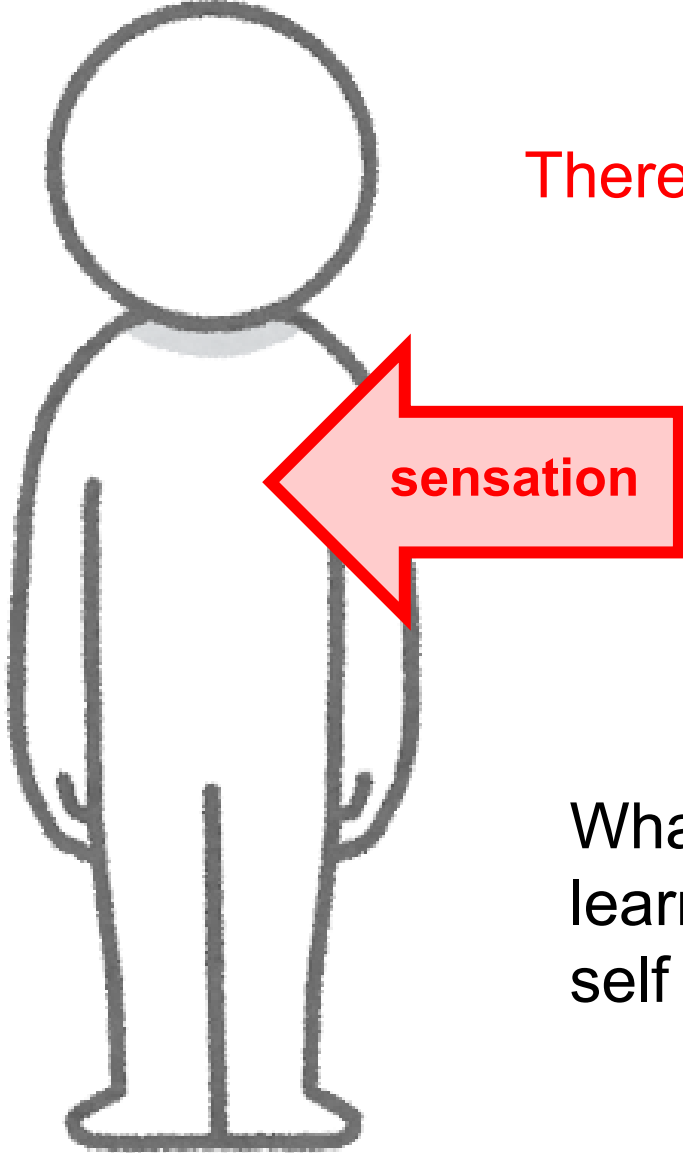


We assume

- there are no pre-existing frames of self and other.
- these concepts themselves are acquired through learning.
- starting from a situation where there is neither self nor other.

How we understand self and others?

There is no distinction between self and other.



sensation

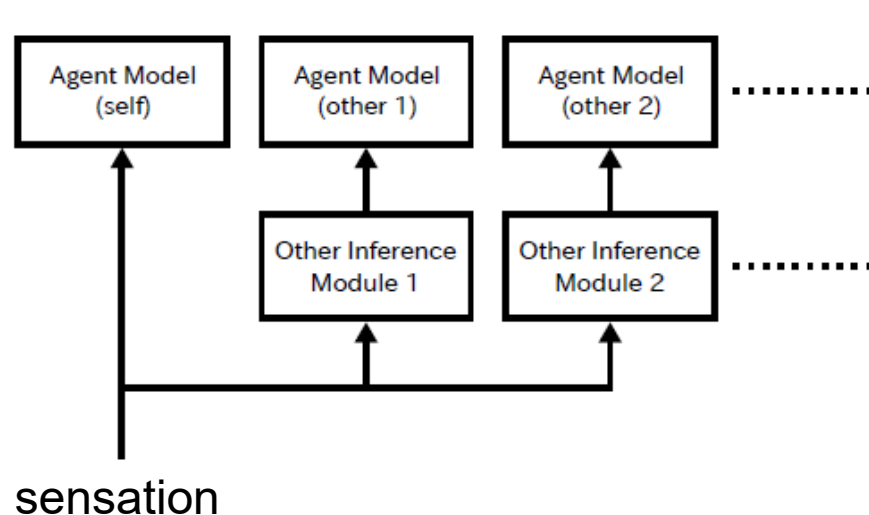
Sensations are **nothing more than external stimuli**, even if they are considered belonging to a particular body from the third-person perspective.

What mechanism makes it possible to learn and obtain the representations of self and other?

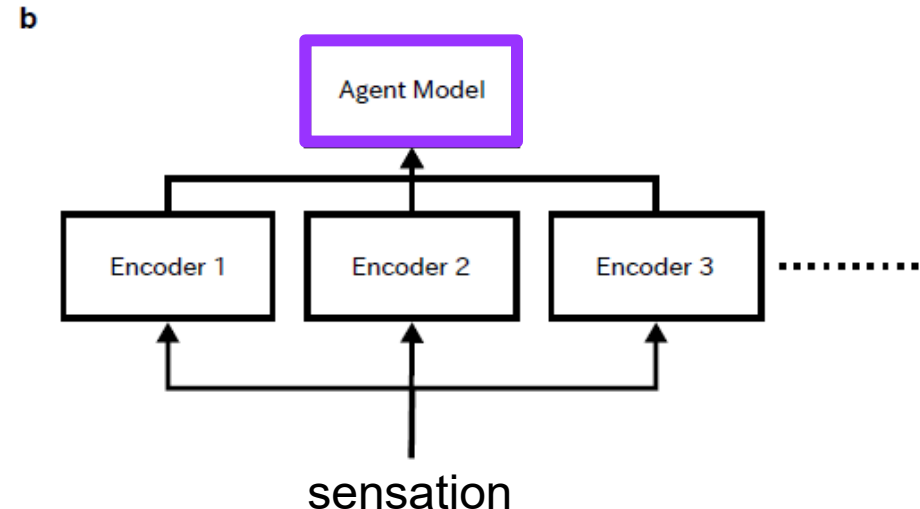
Anyone model that doesn't differentiate between self and others

Instead of creating models of others, we assume a single model for anyone without distinction between self and others.

Building a model of self and others **separately**



Building a **single** model **that doesn't differentiate between self and others**

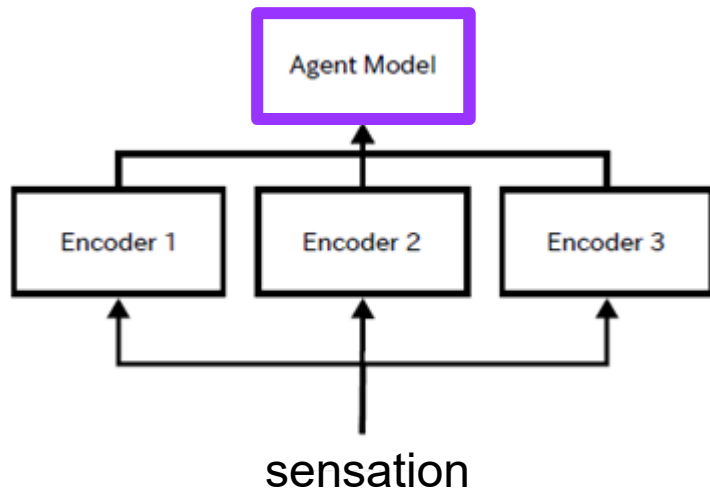


c.f. agent-neutral coding by Gordon (originator of ST)

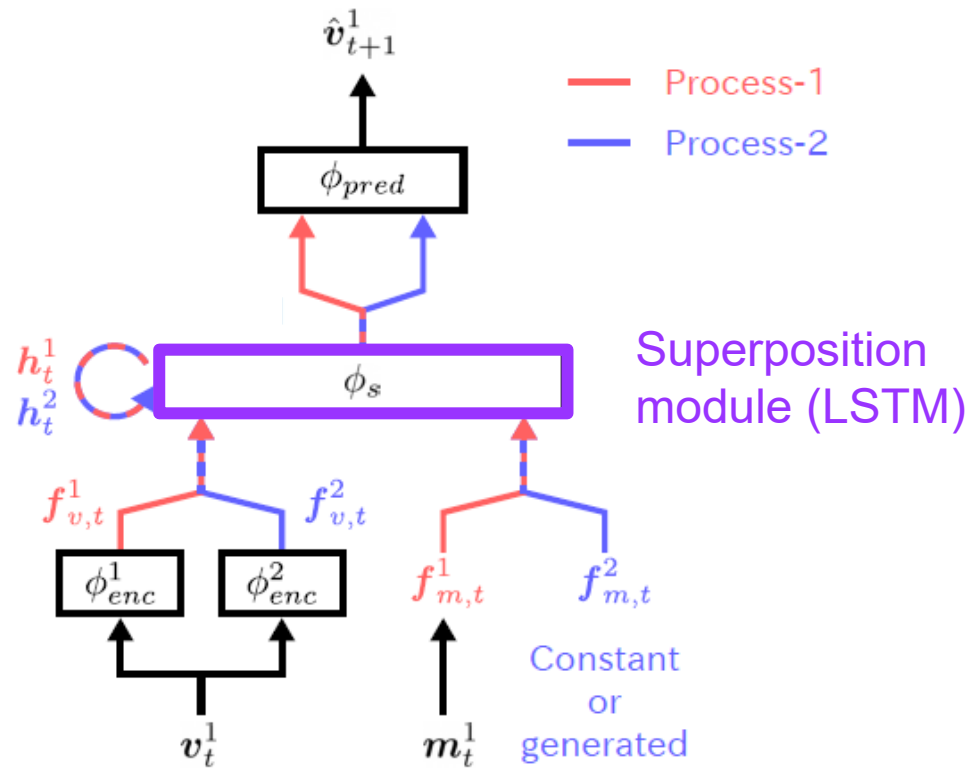
Gordon, R. M. (2021). Simulation, predictive coding, and the shared world. In *The Neural Basis of Mentalizing* (pp. 237-255).

Superposition network model

- Receiving own sensory (visual and somatosensory) inputs
- Multi-stream processing, process-1 and process-2
- Processed by a superposition module
- Integrating outputs of both processes to predict sensory inputs

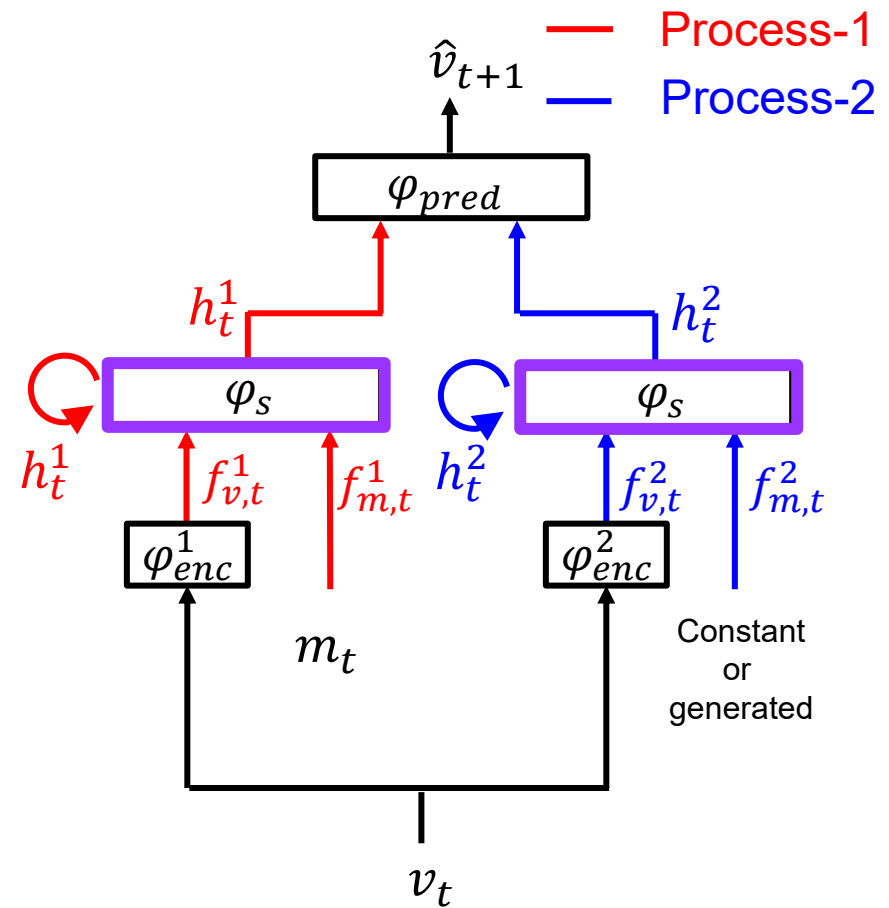
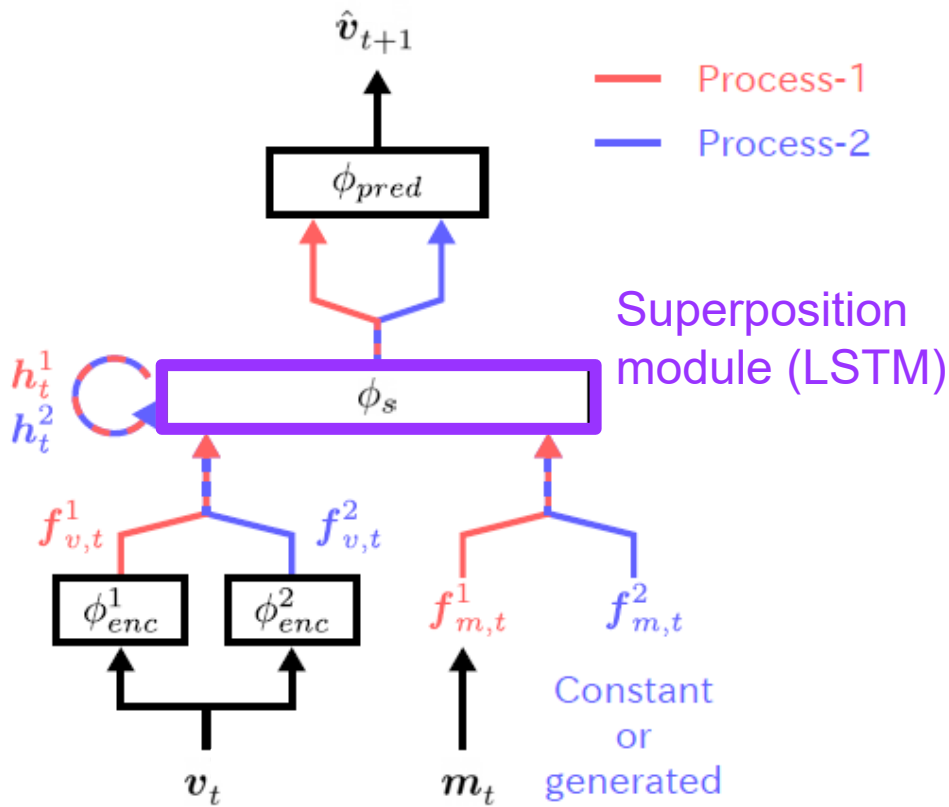


Anyone model (concept)

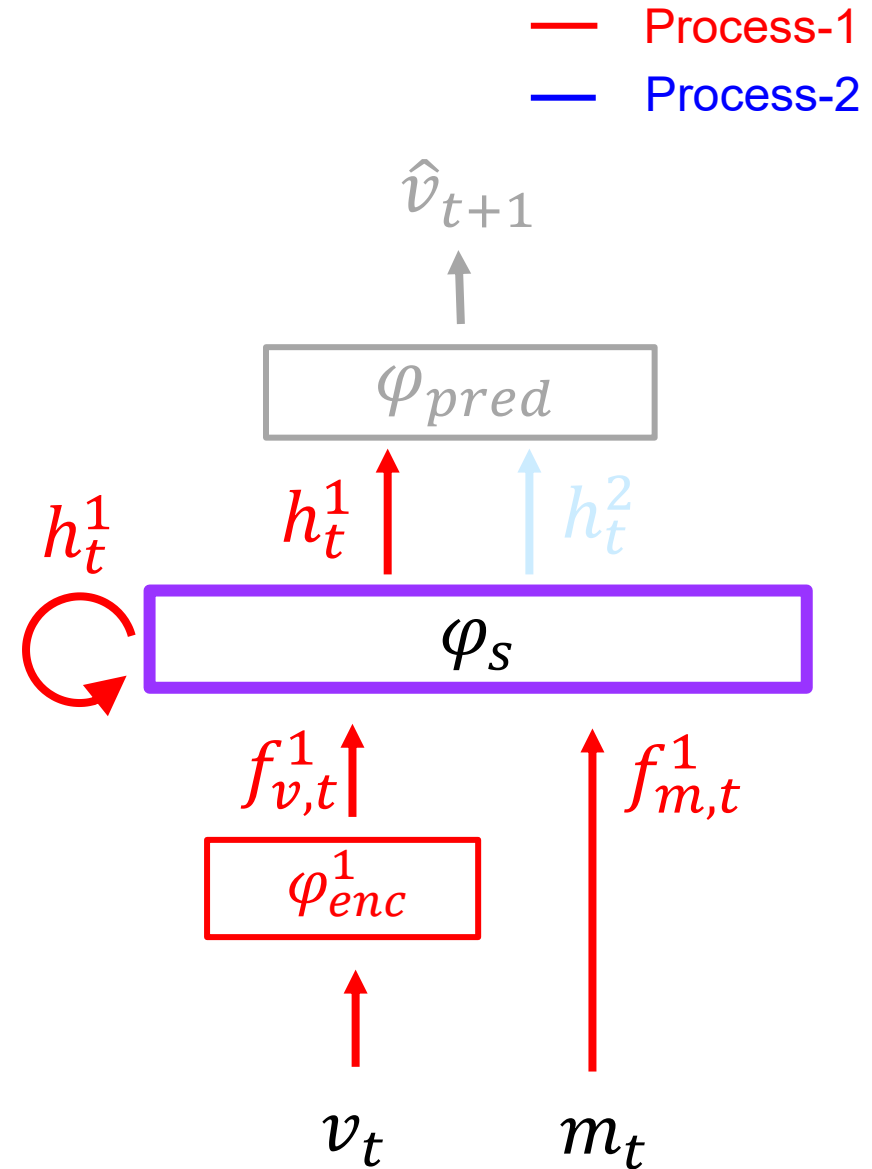
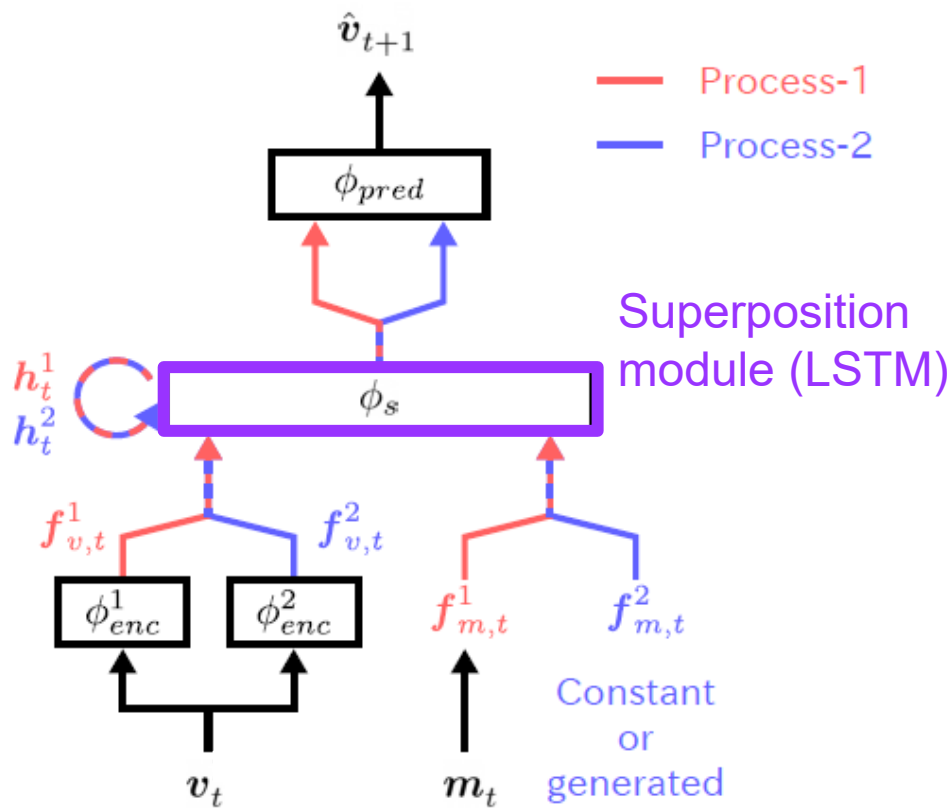


Superposition network model

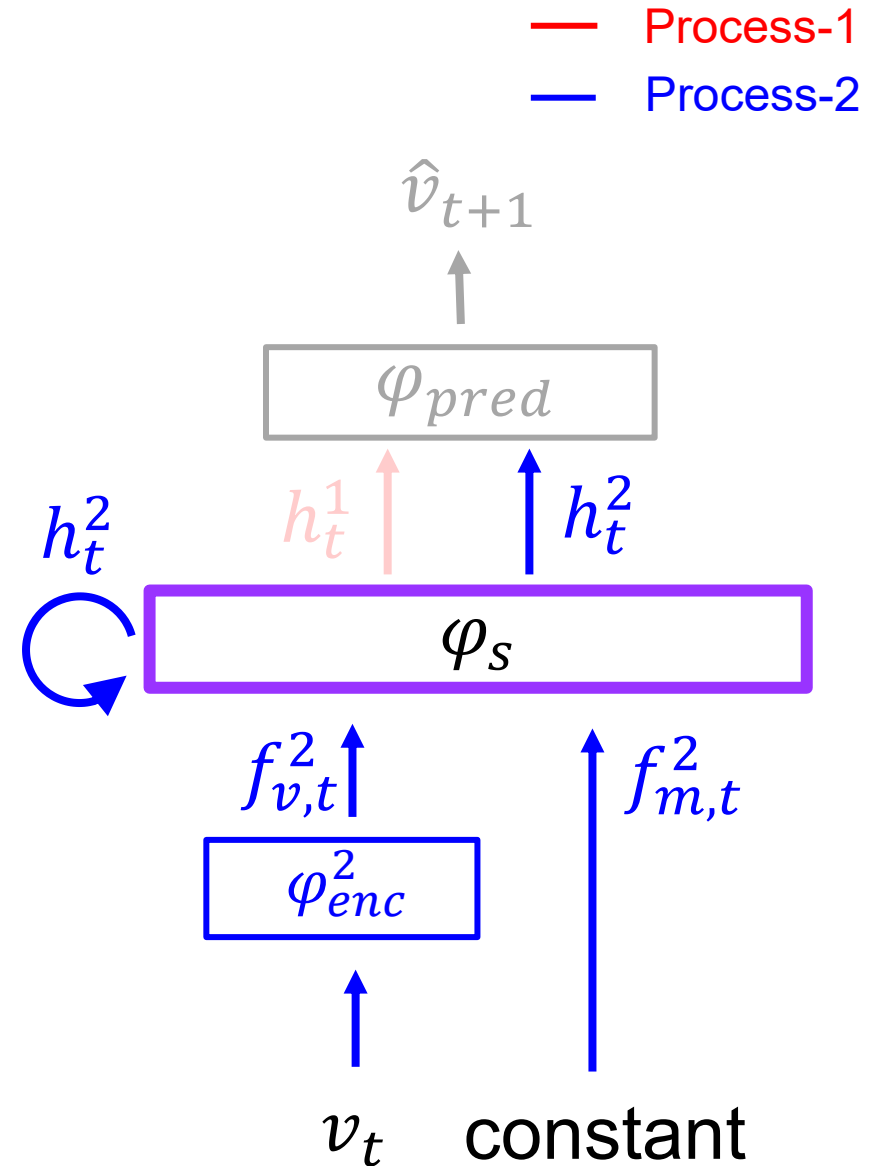
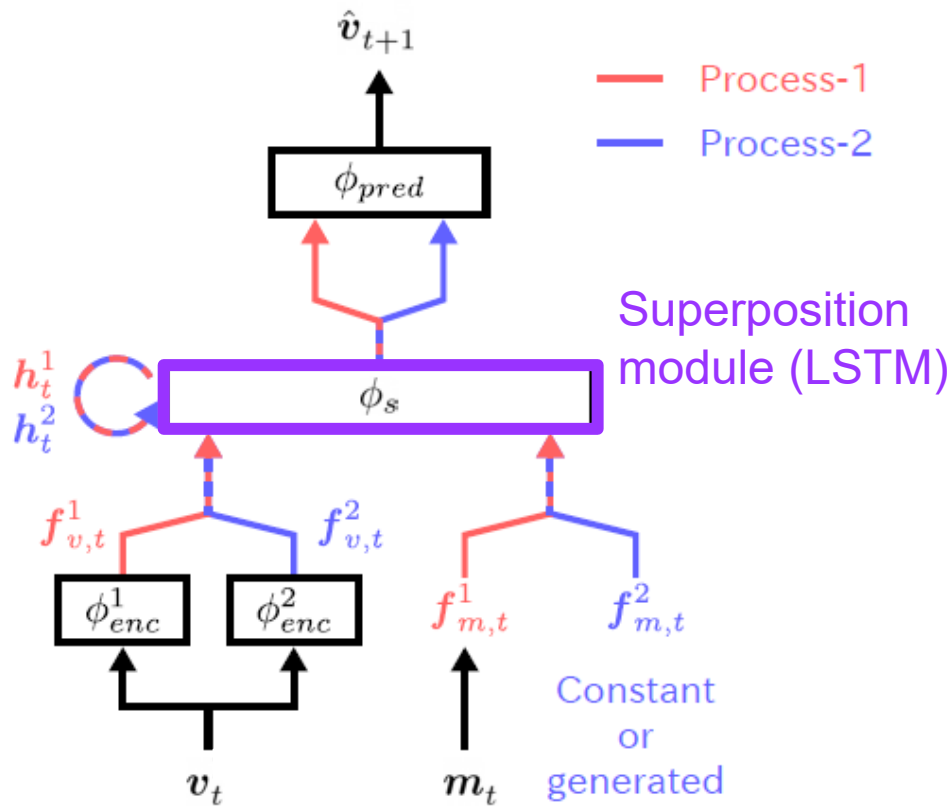
Another illustration form of our model



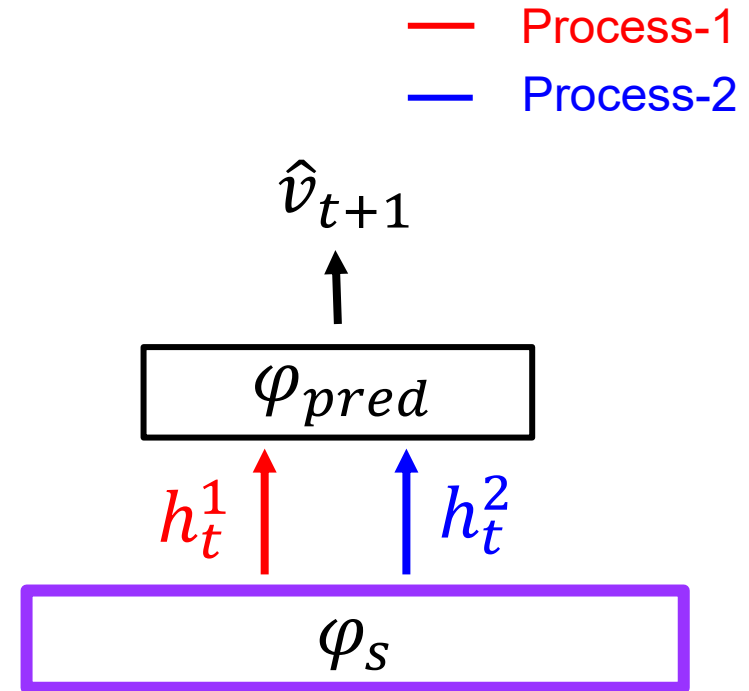
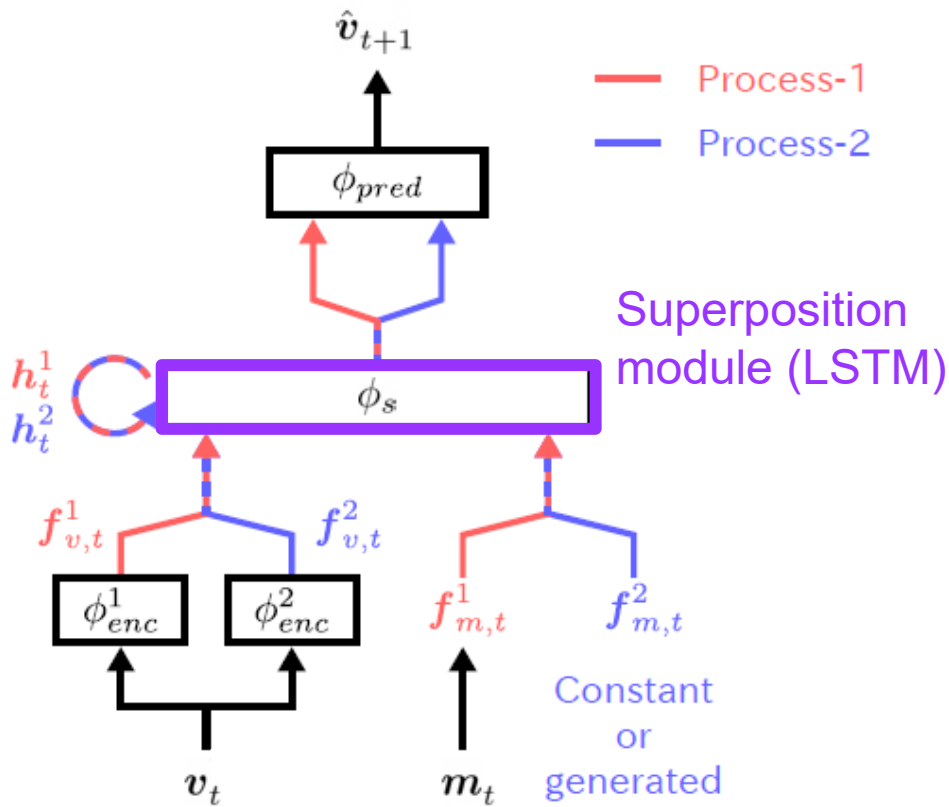
Another illustration form of our model



Another illustration form of our model

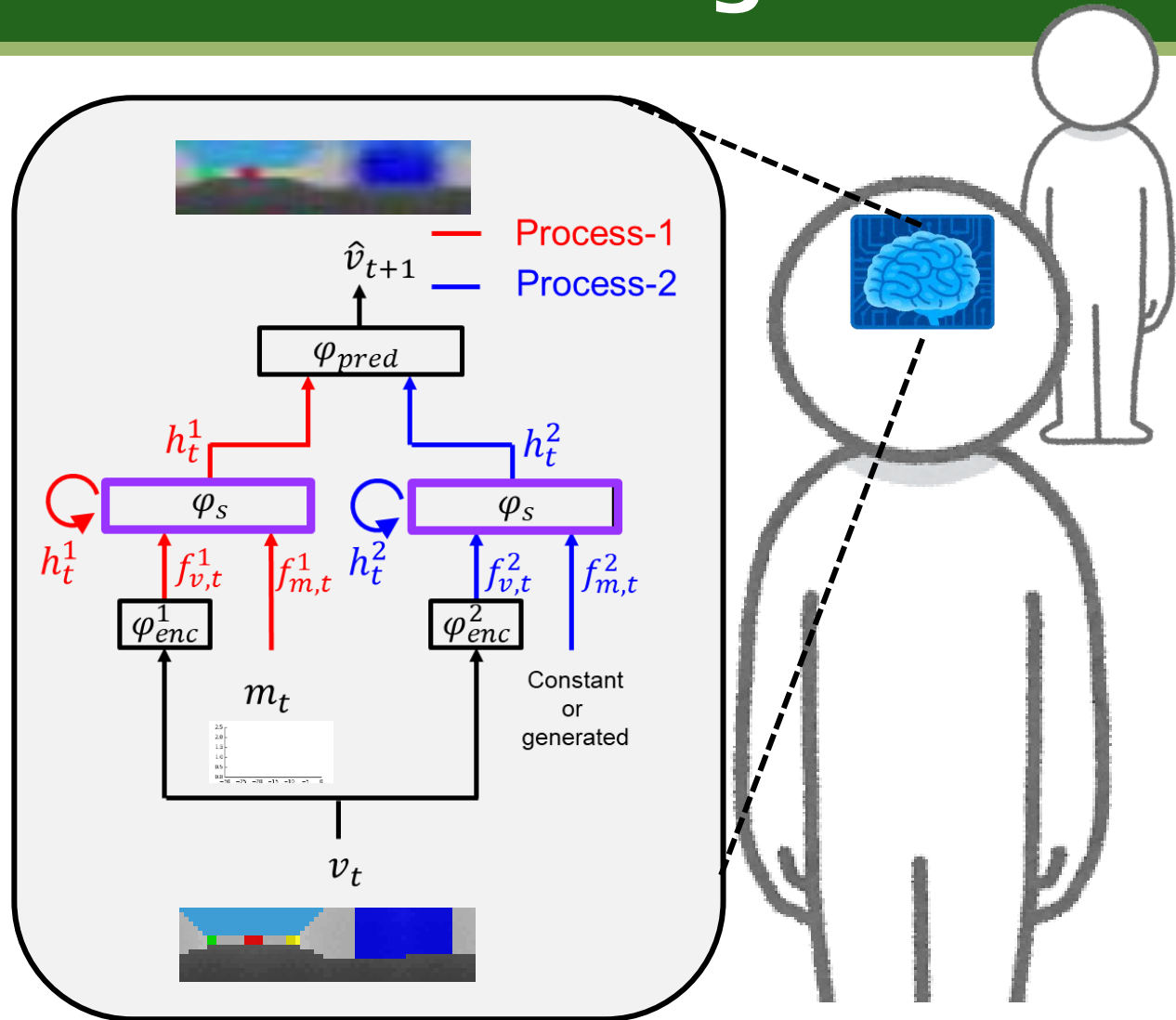


Another illustration form of our model



Predictive learning

The network is trained to minimize the prediction error for vision.



Loss function

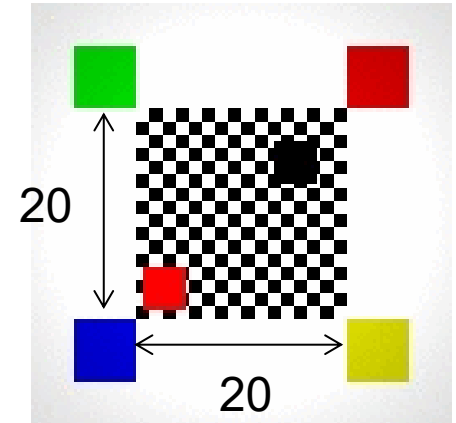
$$\mathcal{L}_{pred} = \sum_{t=1}^T \{ \lambda_{MSE} \text{MSE}(\hat{v}_{t+1}, v_{t+1}) + \lambda_{MAE} \text{MAE}(\hat{v}_{t+1}, v_{t+1}) \},$$

Simulation environment

■ : Agent-1 ■ : Agent-2

Agent-1 and agent-2 exist

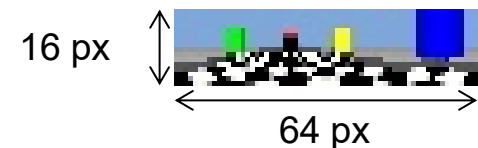
- Agent-1 moves around in the arena.
- Agent-2 does not move but change its locations every trial.



Agent-1:

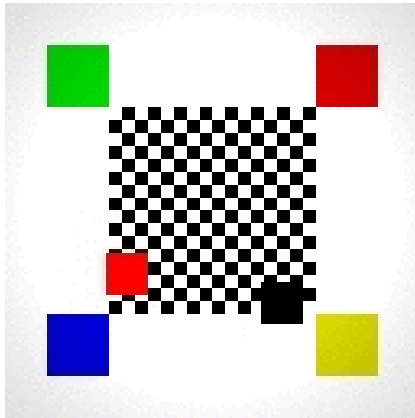
- Equipped with the superposition network
- Receiving visual images and proprioception
- Predicting future visual images

Agent-1's vision



Prediction performance

Example of agent-1's trajectory



visual prediction

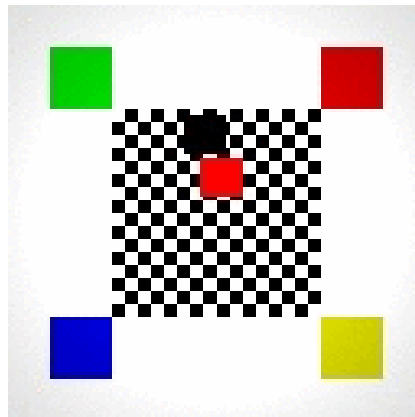
Truth



predict



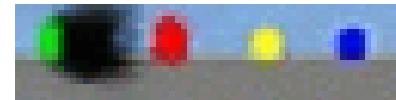
Another example



Truth

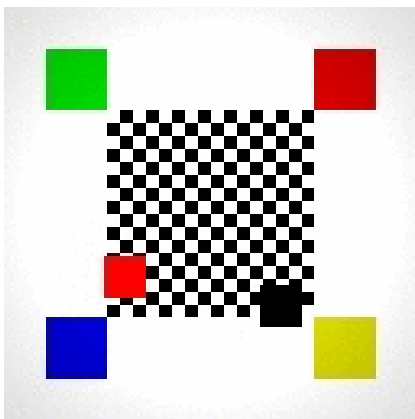


predict

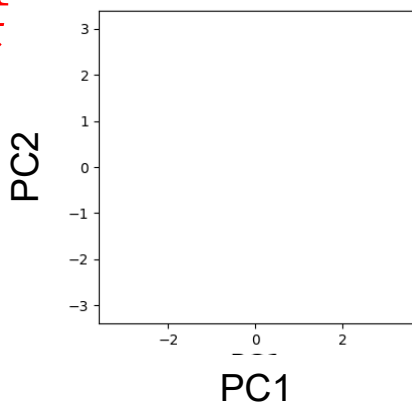


Visualizing internal states of SM

Example of agent-1's trajectory

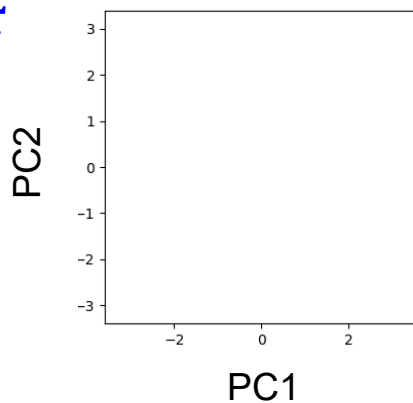


h_t^1

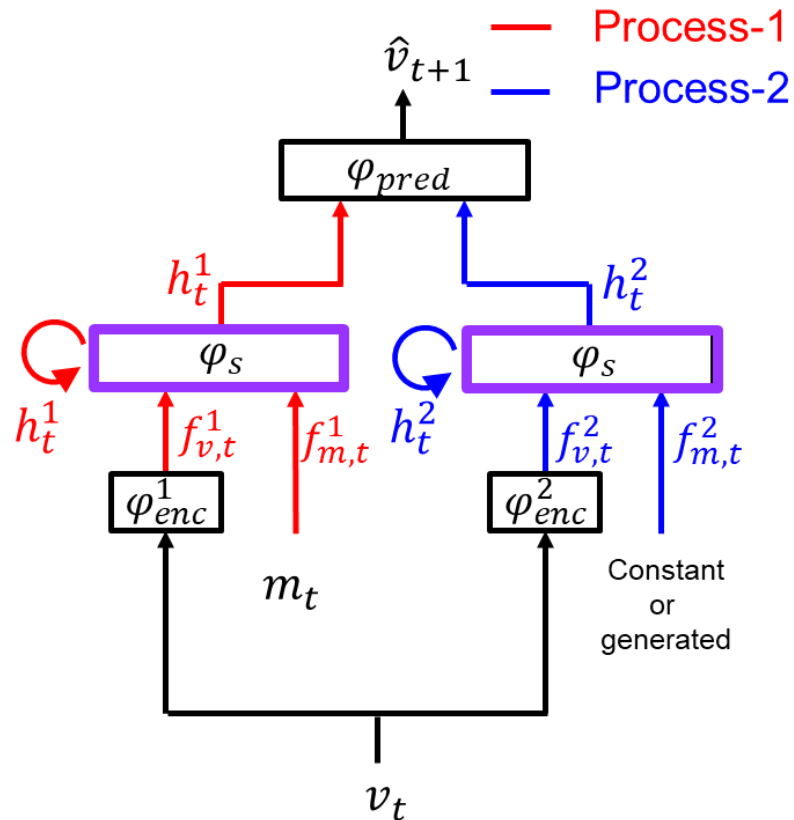


Coloring : agent-1's positions

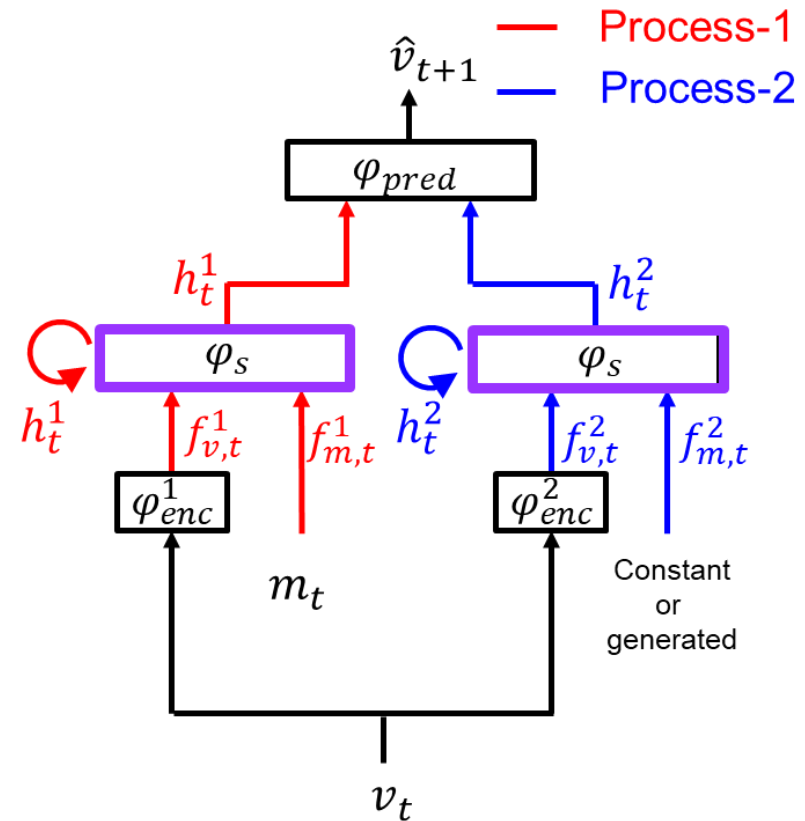
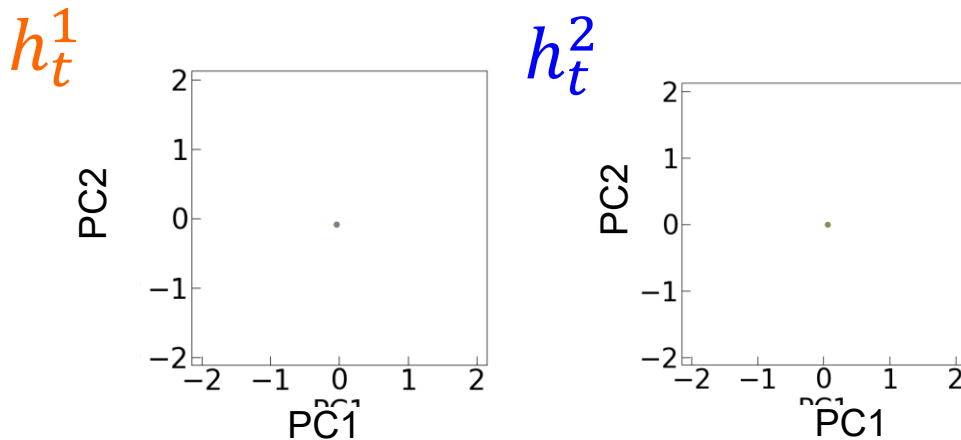
h_t^2



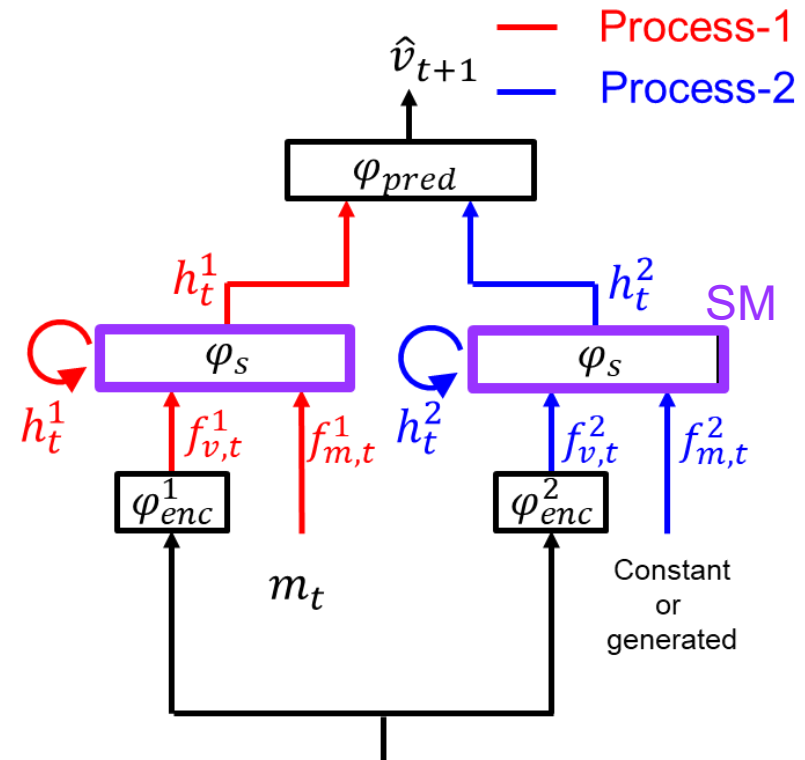
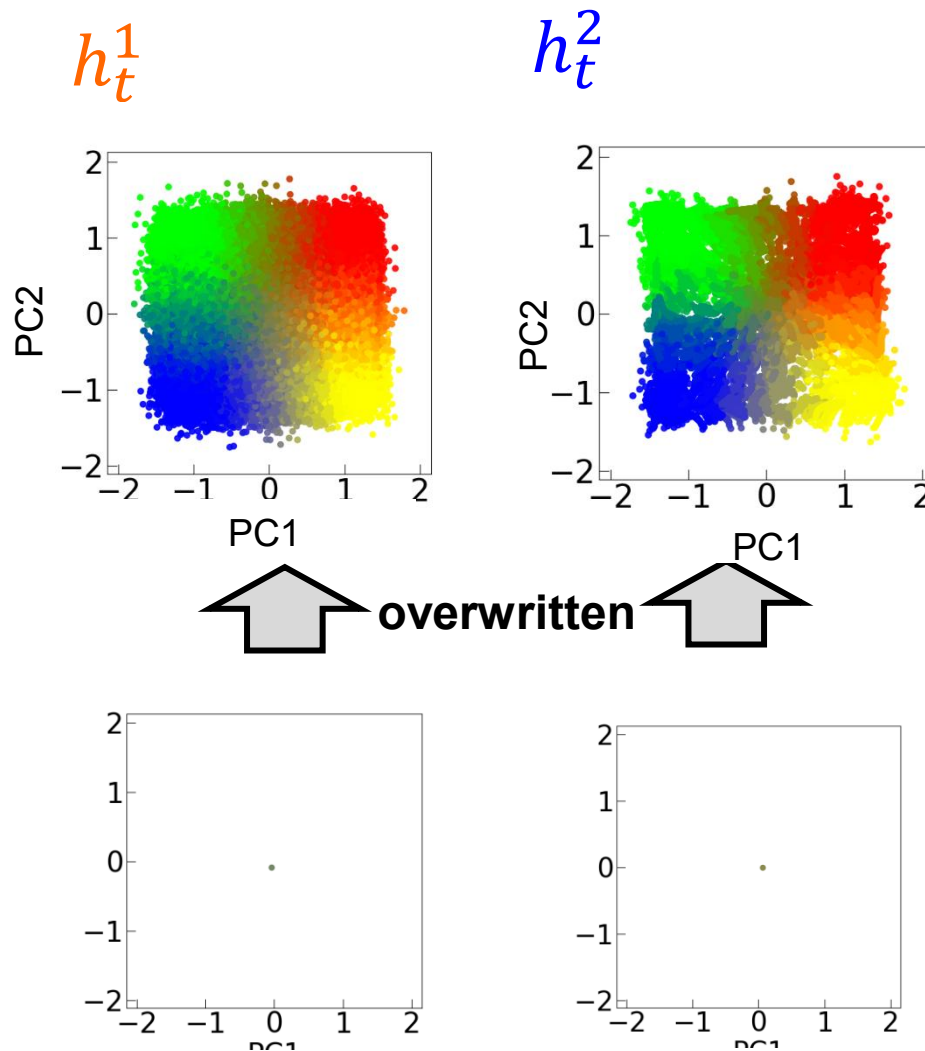
agent-2's positions



Visualizing internal states of SM



Visualizing internal states of SM



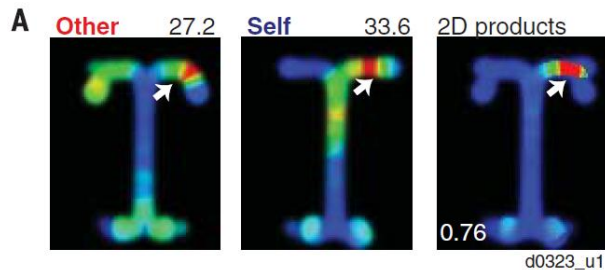
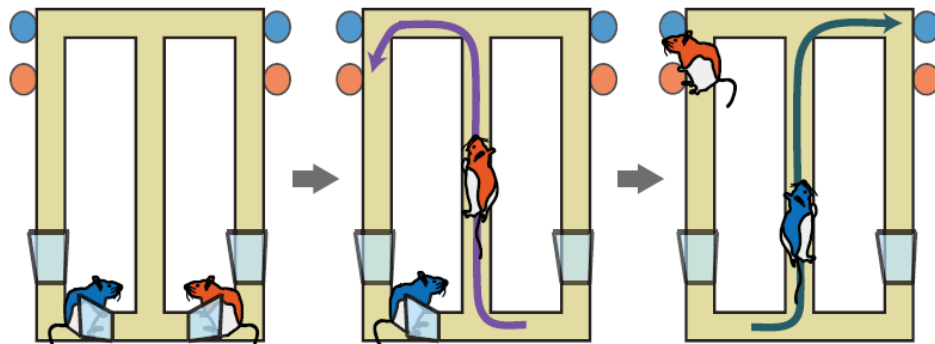
- The internal states of SM are activated in response to A-1 and A-2's positions.
- Similar to the social place cell found in neuroscience.

[Danjo, et. al, 2018, Omer, et. al, 2018]

Discovery of Social Place Cells

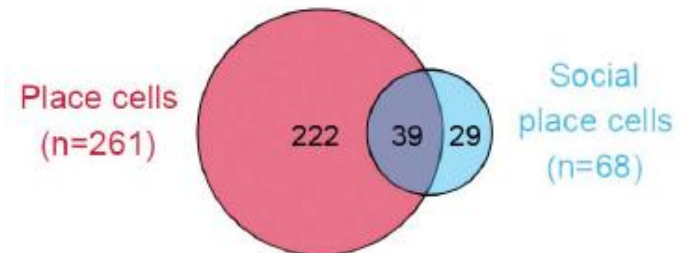
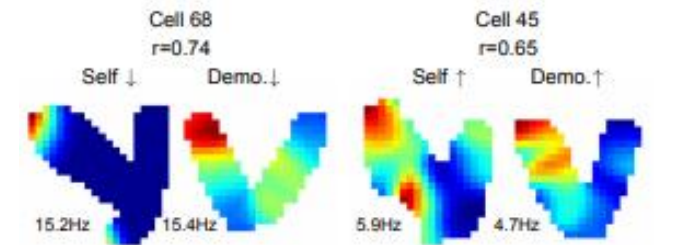
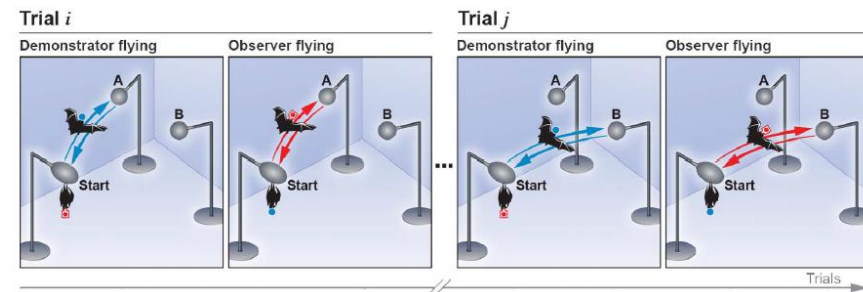
Spatial representations of self and other in the hippocampus

[Danjo et al., 2018]



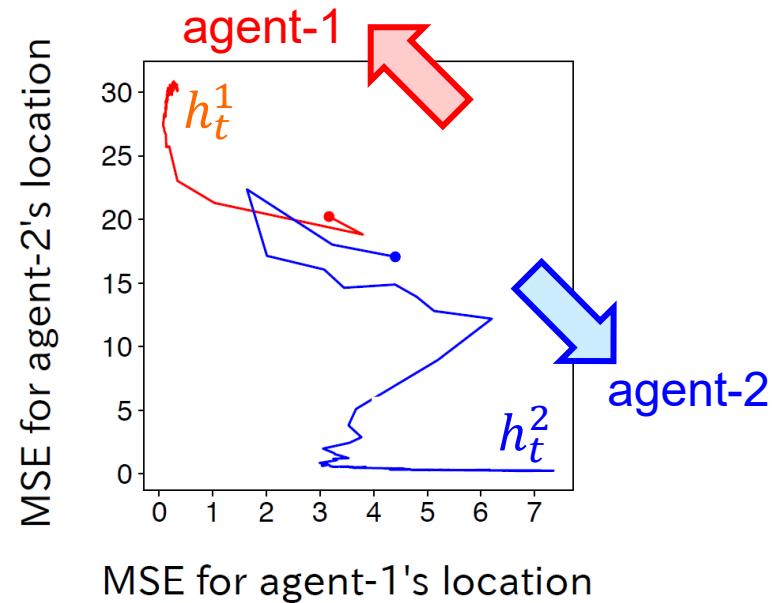
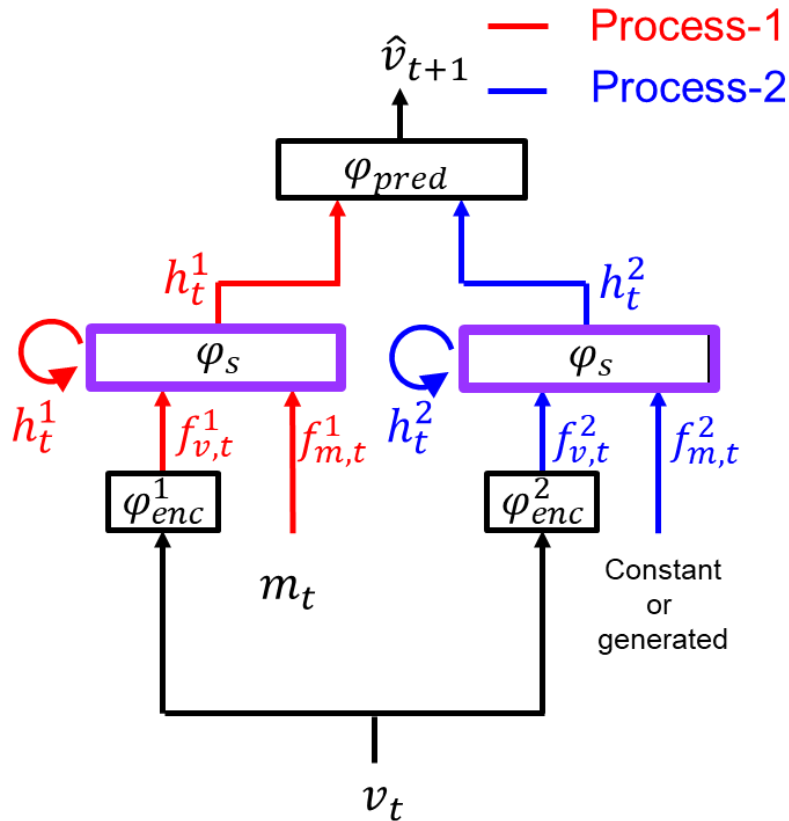
Social place-cells in the bat hippocampus

[Omer et al. 2018]



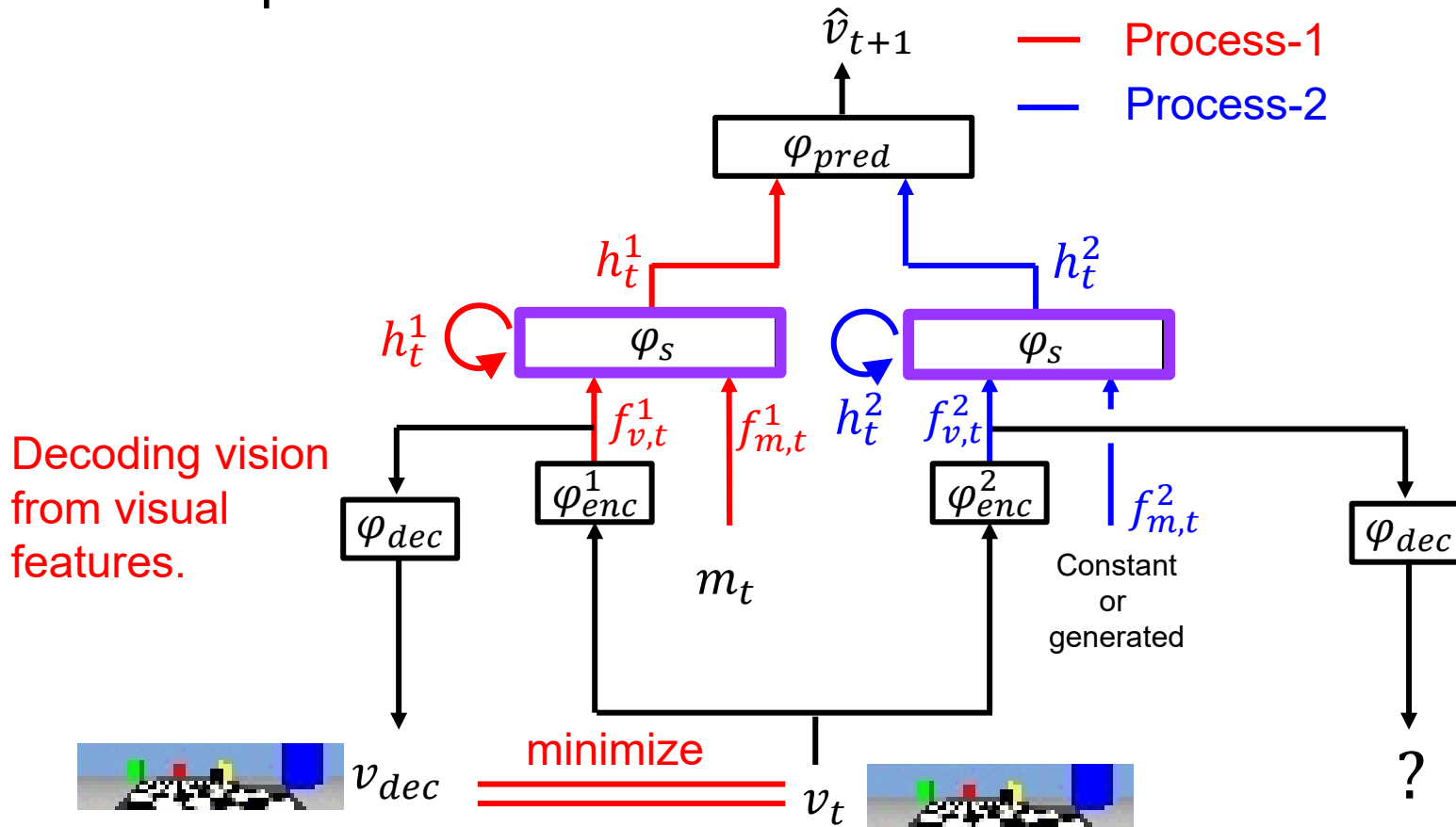
Self / other separation

Process-1 and -2 are gradually specialized to self and other, respectively. Process-1 handles information about self and process-2 handles information about other.



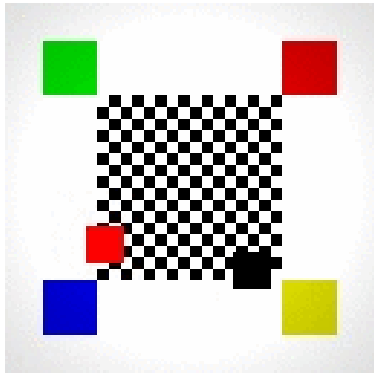
Visualizing what agent-1 sees

- Building a decoder to reconstruct the visual image from (own) visual features on process-1.
- Applying the decoder to the visual features on process-2 and visualize what it sees.



Visual perspective taking

Example 1

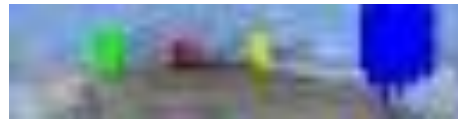


Process-1

A-1's
vision



Decode

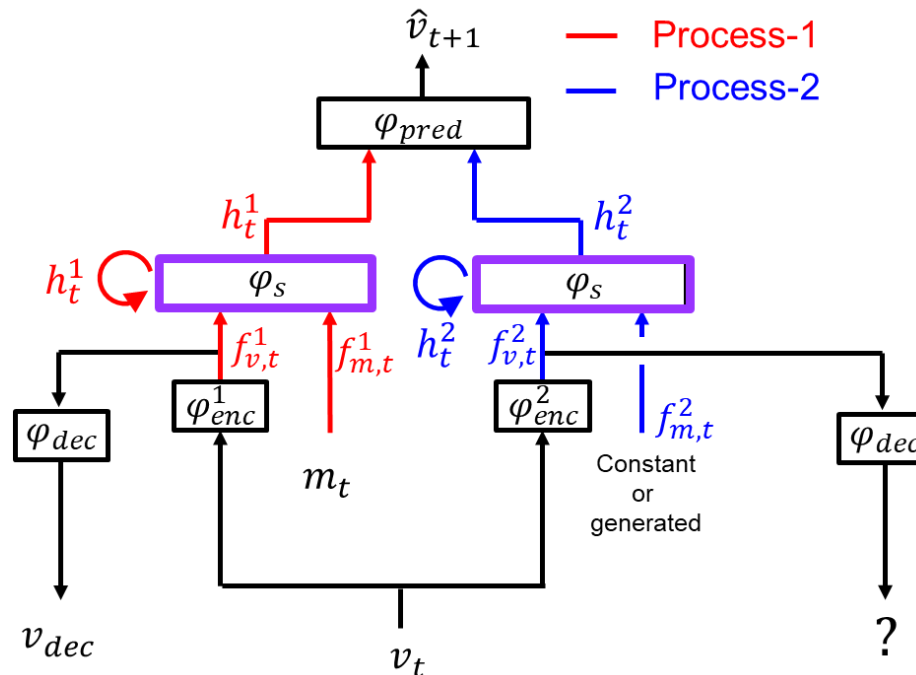
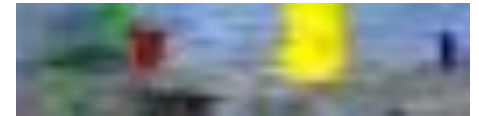


Process-2

A-2's
vision

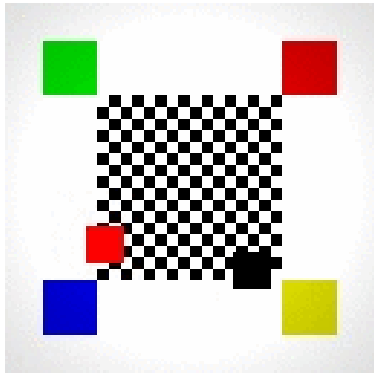


Decode



Visual perspective taking

Example 1

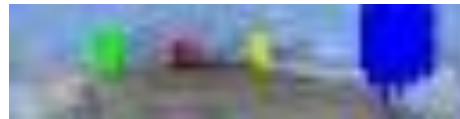


Process-1

A-1's
vision



Decode

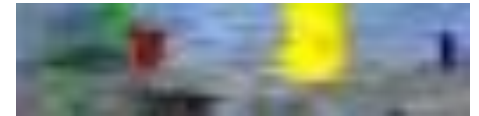


Process-2

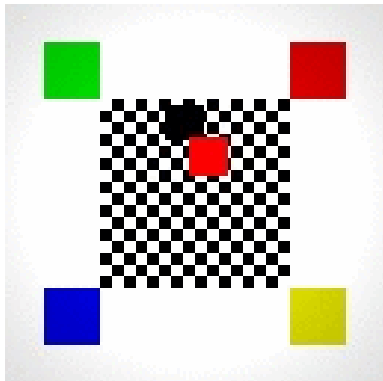
A-2's
vision



Decode



Example 2

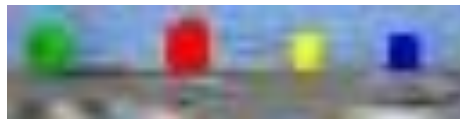


Process-1

A-1's
vision



Decode

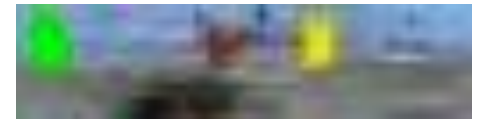


Process-2

A-2's
vision



Decode



Just by using the superposition module, the agent can naturally obtain **other's perspective**.

Training motion generator

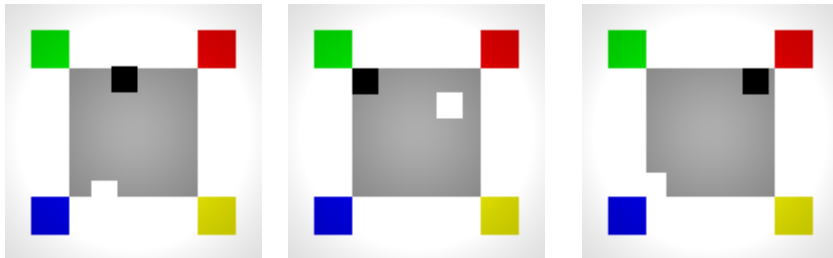
■ Agent-2 has intentions and begins to move

(So far, only agent-1 was movable)

- Agent-2 moving around
- Adding a **motion generator (MG(LSTM))**
- Training the MG via predictive learning

■ Three motion patterns of Agent-2

(Agent-1 : **Moving in a random way same as before**)

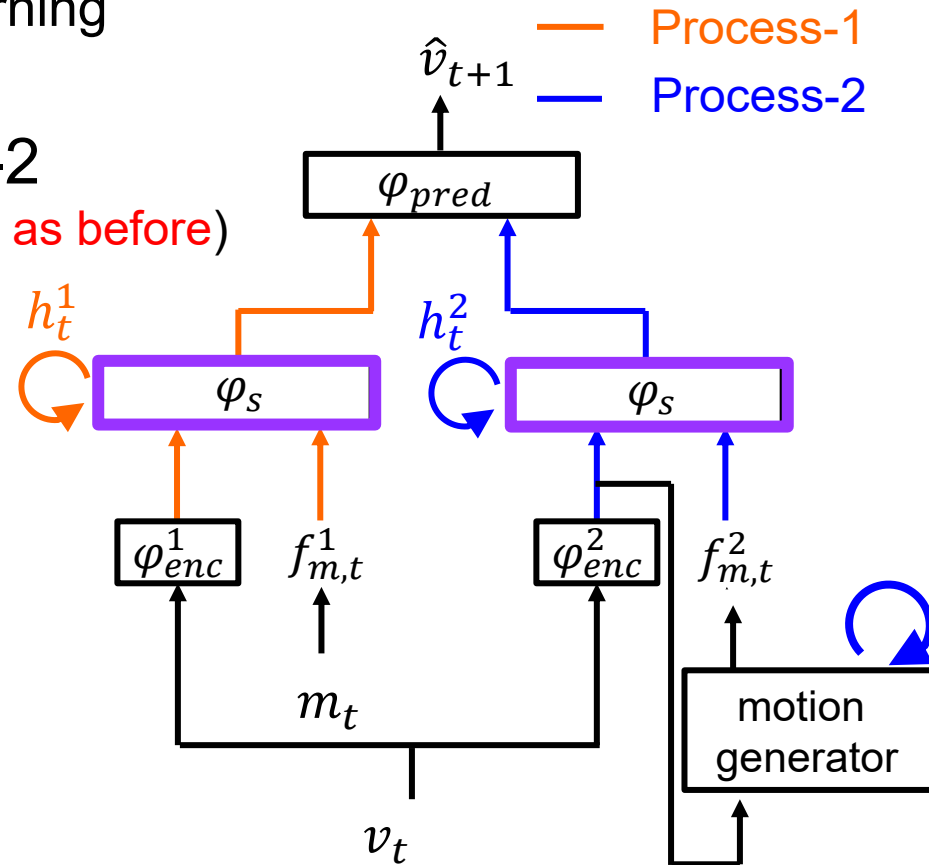


Stopping

CW

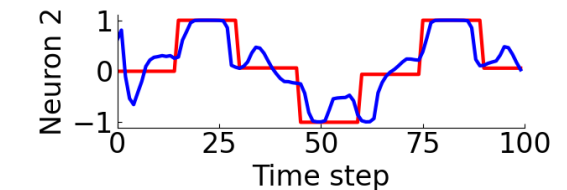
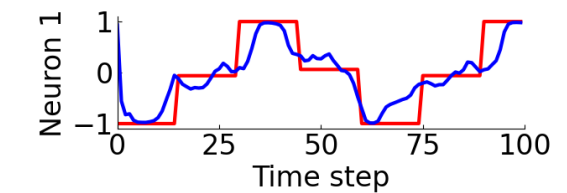
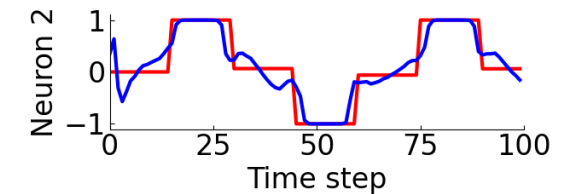
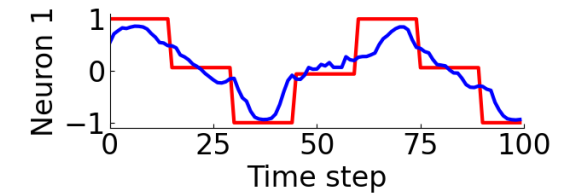
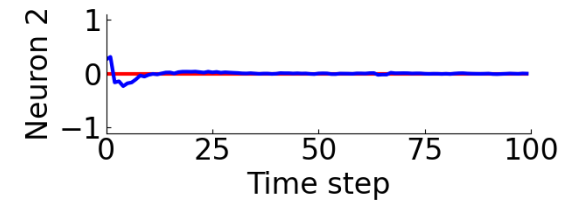
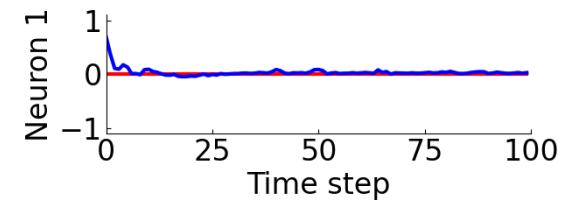
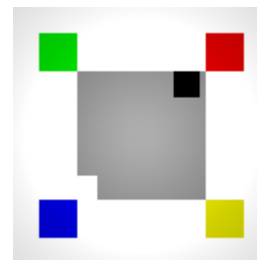
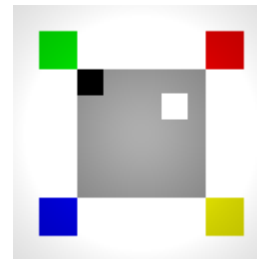
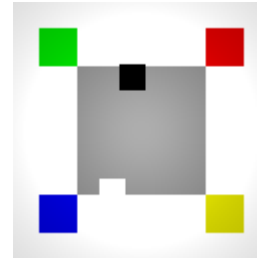
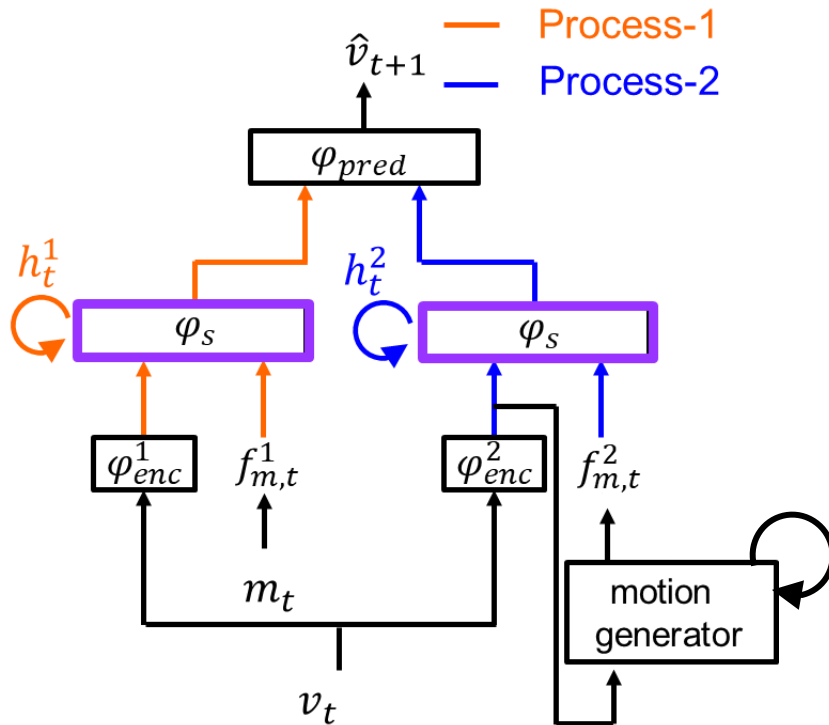
CCW

White : Agent-1 Black : Agent-2



Mirror neuron-like activations

— Motion commands that create A-2's movements as if A-1 were A-2.

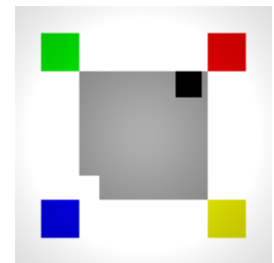
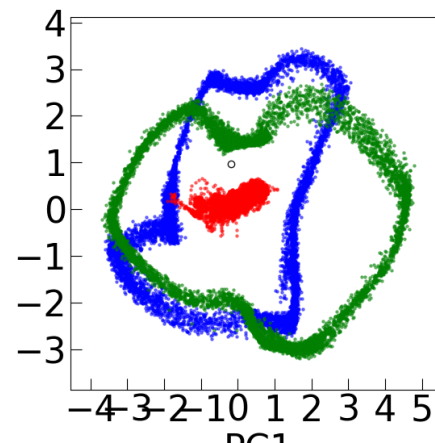
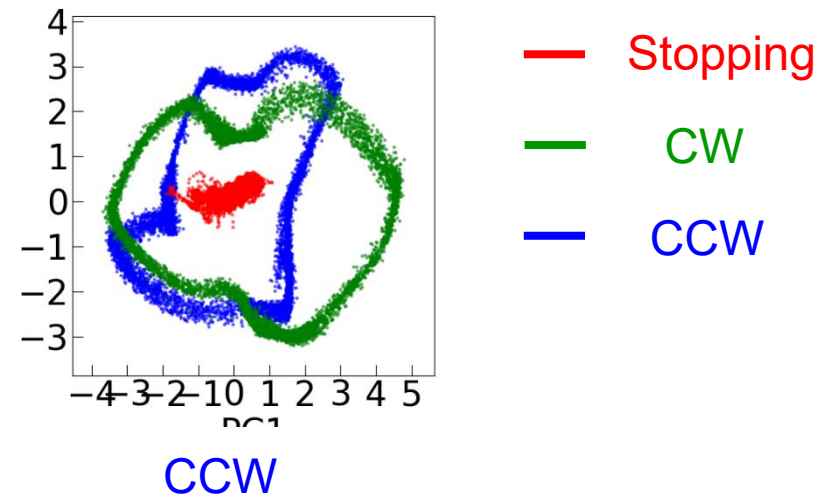
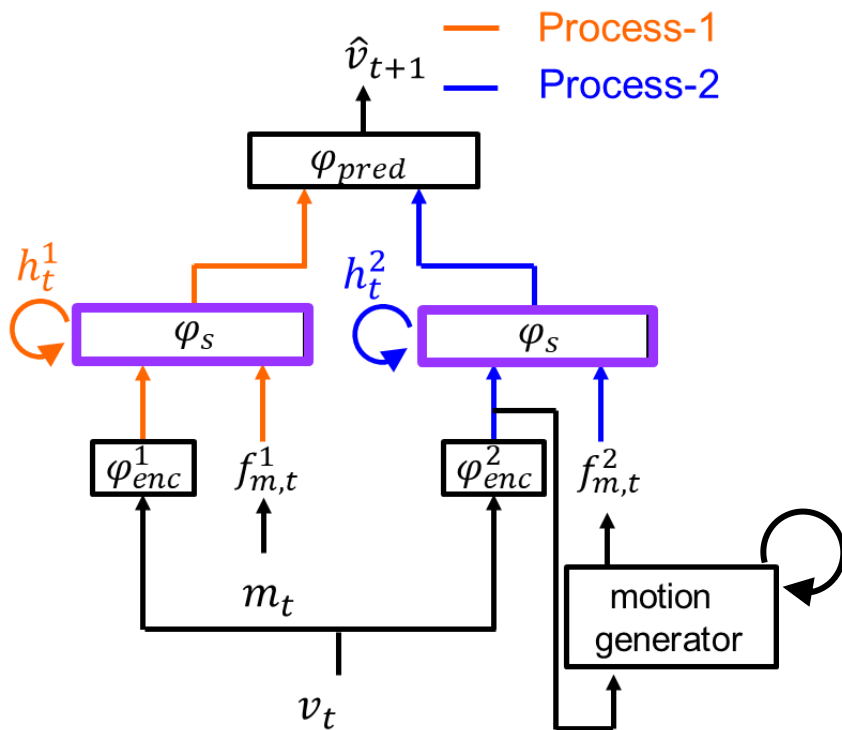


- When observing agent-2, motions to generate agent-2's movements are internally created within agent-1.

White : Agent-1
Black : Agent-2

Internal states of motion generator as intentions of the other

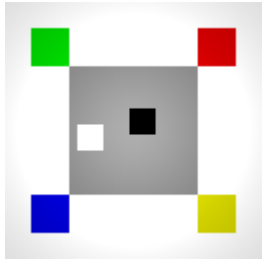
- Visualizing the internal states of the motion generator by color-coding them for each action of Agent-2.



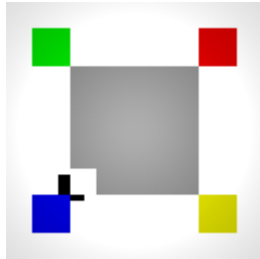
As a result, the attractors corresponding to the behavior patterns were formed.

Using MG of the other as self's

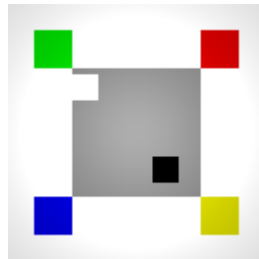
- Although these are patterns agent-1 has never moved in before, agent-1 tries moving in the three patterns



Stopping



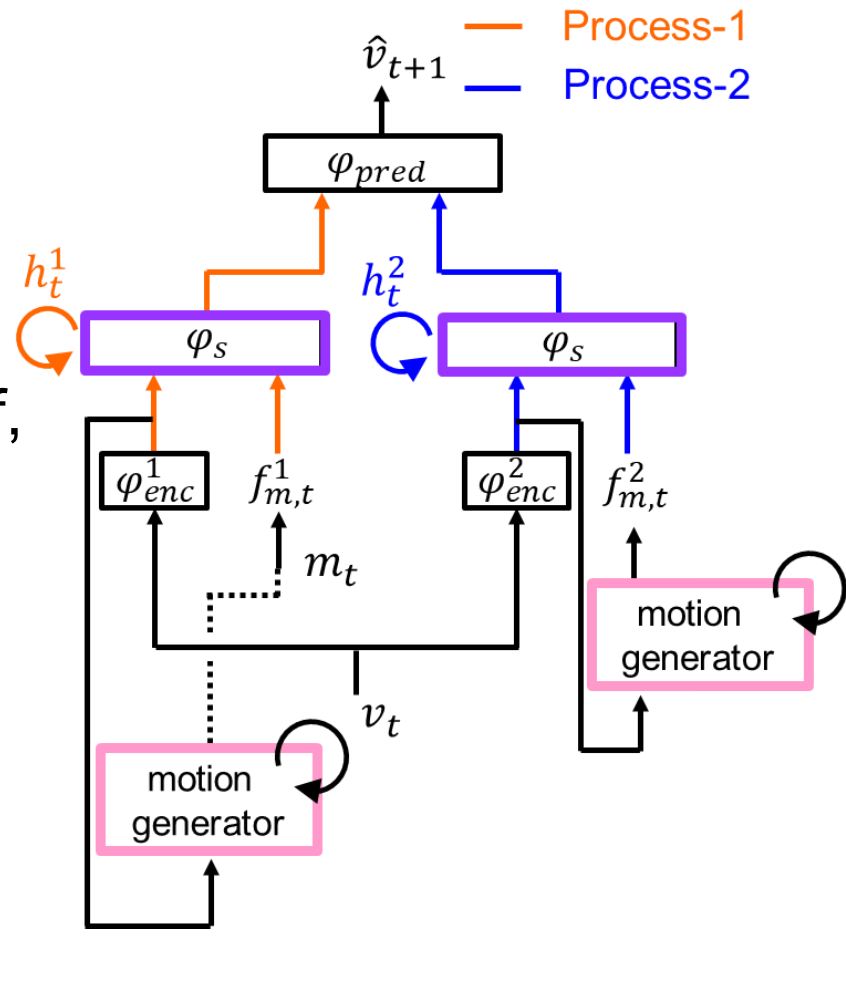
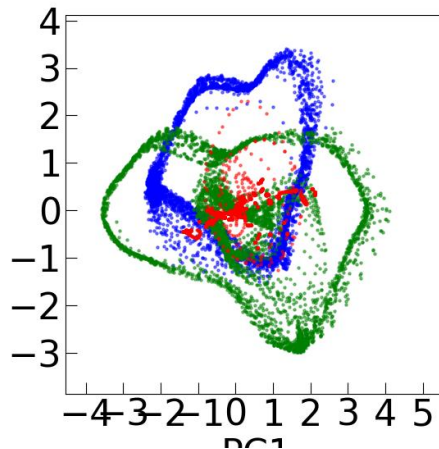
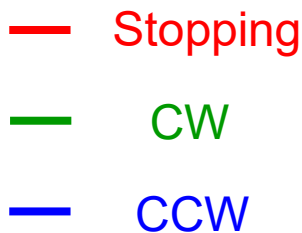
CW



CCW

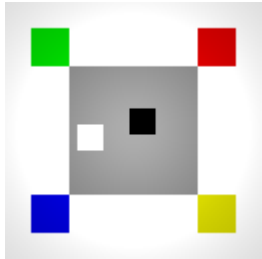
White : Agent-1 Black : Agent-2

- Applying the MG of the other to self, visualizing the internal states same as before.

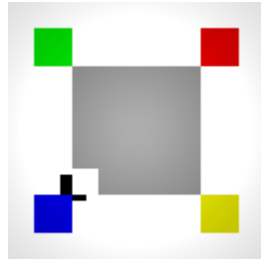


Using MG of the other as self's

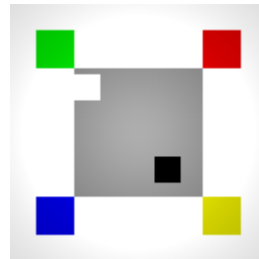
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Stopping



CW

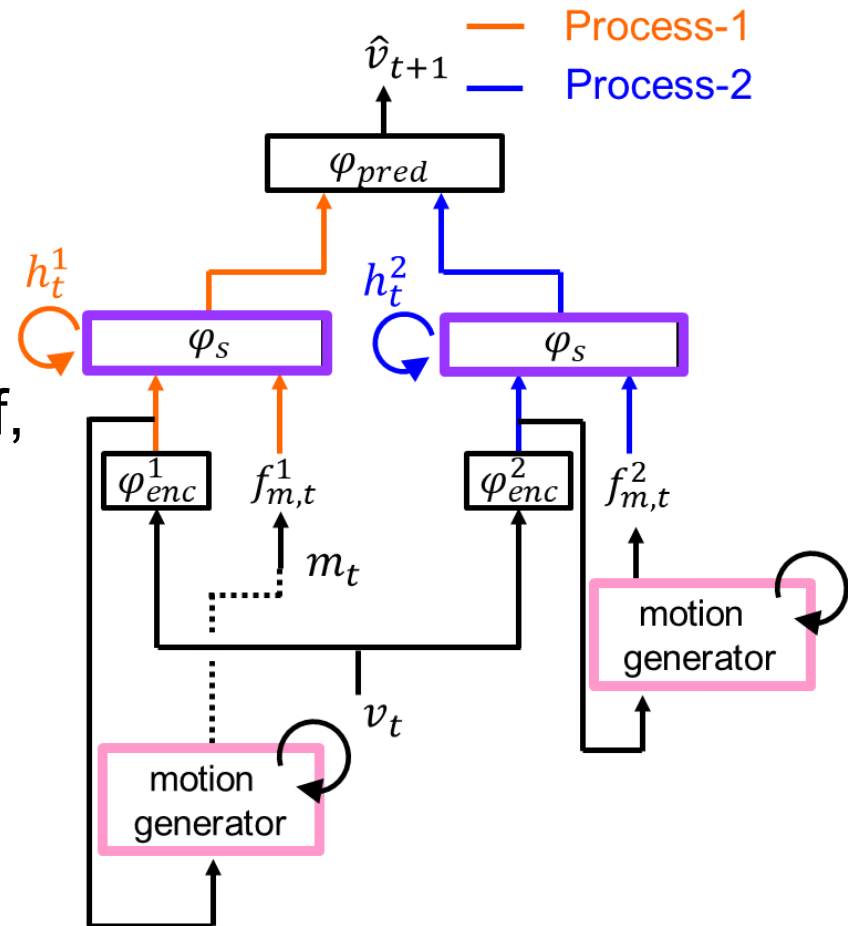
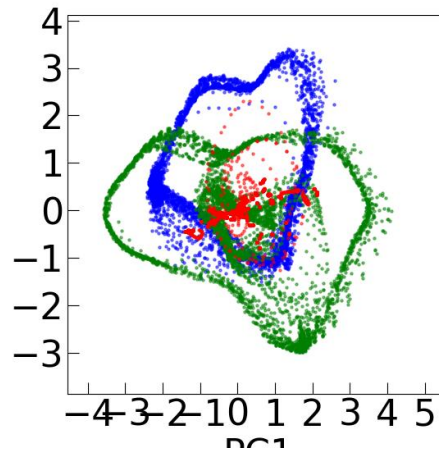


CCW

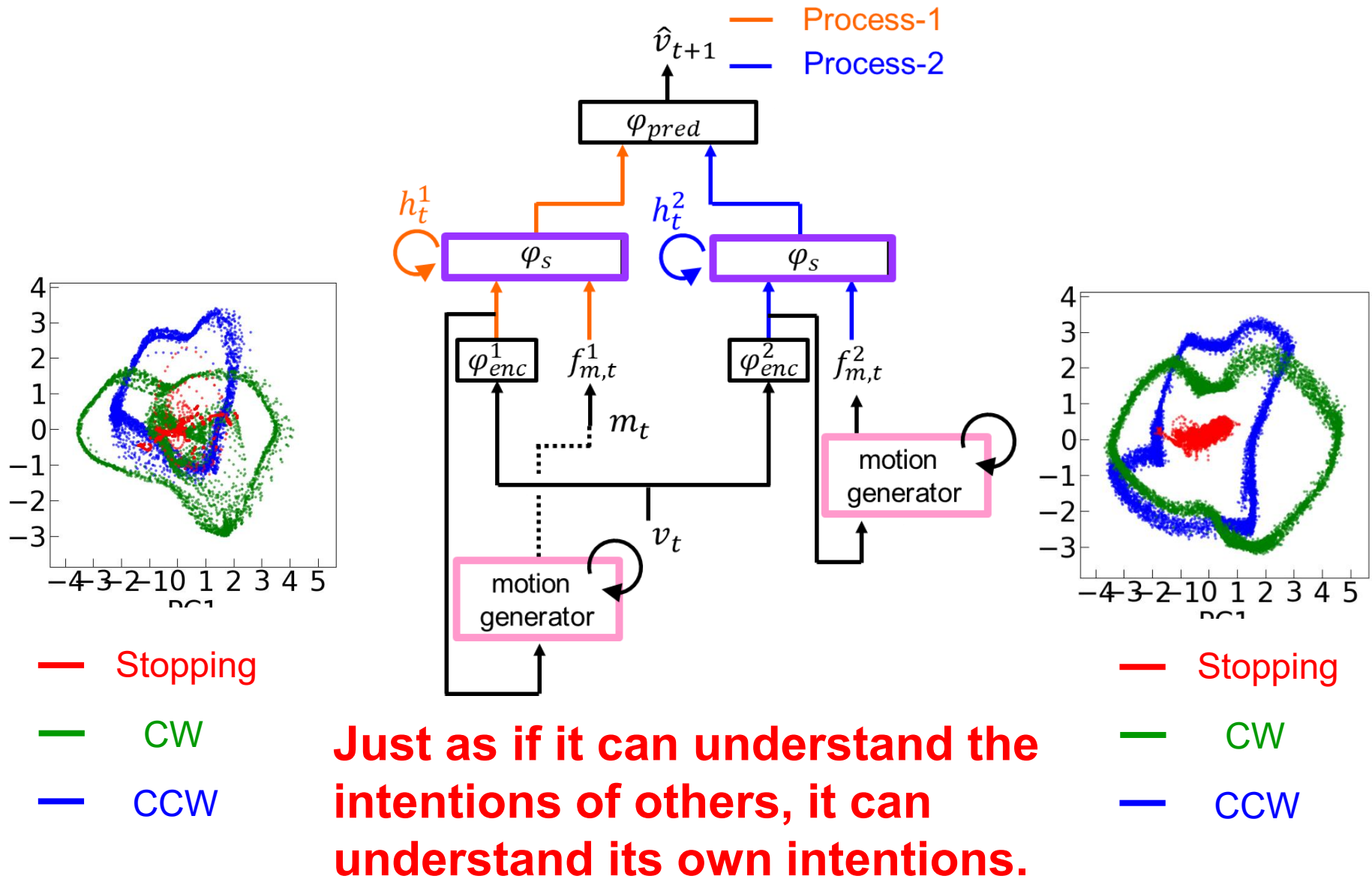
White : Agent-1 Black : Agent-2

- Applying the MG of the other to self, visualizing the internal states same as before.

— Stopping
— CW
— CCW



Comparison between self and other

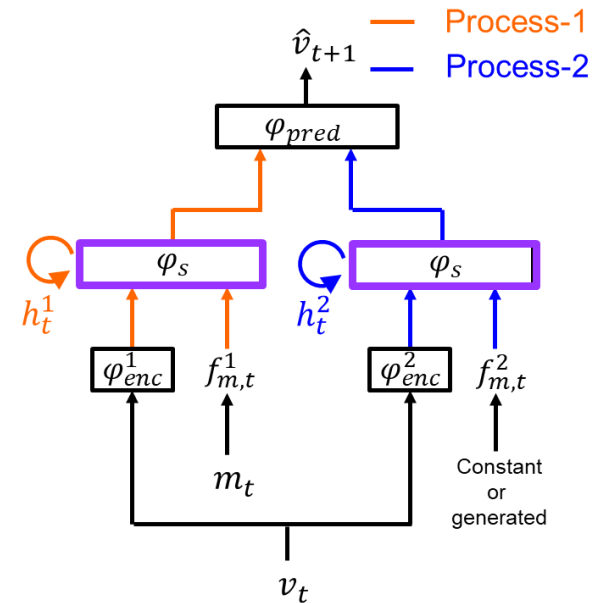


What we found

The superposition mechanism

- No pre-given frameworks of self and other
- Self / other differentiation
- Shared representation (or social place cell)
- Visual perspective taking
- Mirror neuron like activations
- Forming one's own intention through other's intention

These abilities are acquired simply via predictive learning.



Predictive learning of sensory inputs

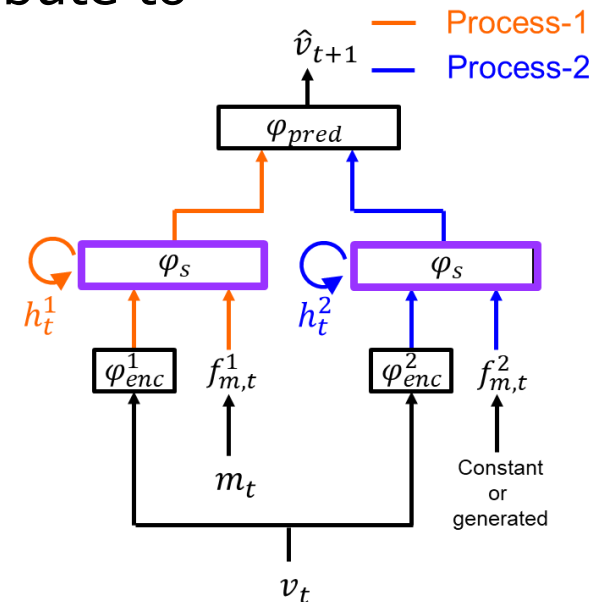
- Self-supervised learning
- Rich learning resources and costless for infants
- Predictive processing

Superposition mechanism exists in the brain?

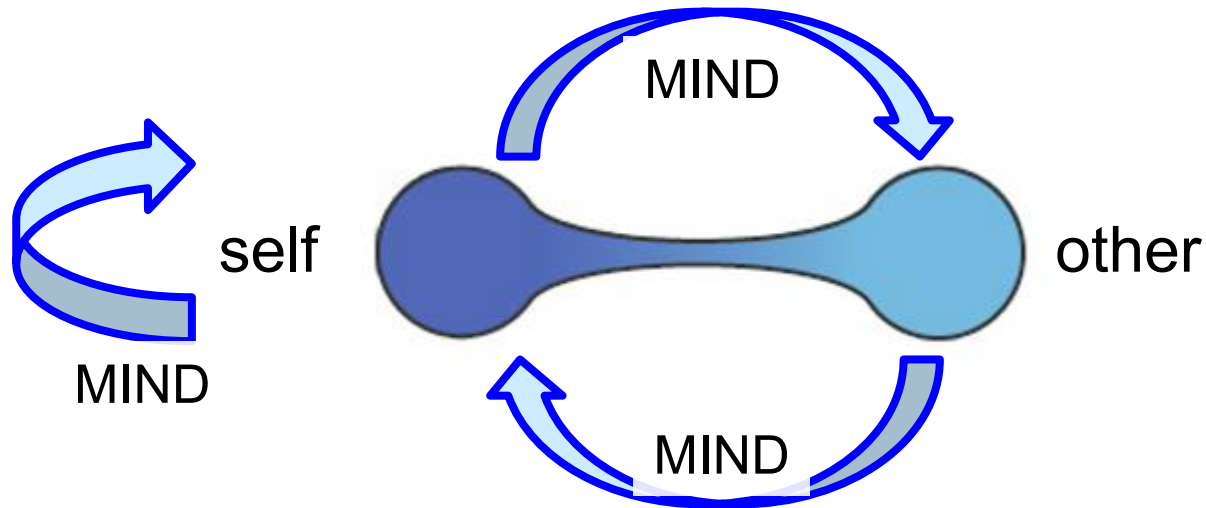
Neuroscience evidences

- Many evidences for the existence of shared activations for self-experiences and empathic experiences
- rTPJ responsible for self/other distinction and visual perspective-taking.
- Whether or how the shared activations contribute to the social cognition is still controversial.

If the superposition mechanism exists in the brain, the shared activations can be developed and basic abilities to support social cognition can be acquired naturally.



Mind and agency comes from others



Superposition mechanism (shared module network)

❑ Multisensory integration model using superposition

- Wataru Noguchi, Hiroyuki Iizuka & Masahito Yamamoto (2022) Multi-modal shared module that enables the bottom-up formation of map representation and top-down map reading, *Advanced Robotics*, 36:1-2, 85-99
- Wataru Noguchi, Hiroyuki Iizuka & Masahito Yamamoto (2019). Navigation behavior based on self-organized spatial representation in hierarchical recurrent neural network, *Advanced Robotics*, 33:11, 539-549

❑ Symbol grounding problem

- Yasuhiro Shimada, Wataru Noguchi, Hiroyuki Iizuka, Masahito Yamamoto, Bottom-up formation of number representation and top-down understanding of symbolic manipulation, *Proceedings of the ALIFE 2022*, pp. 337-344 (2022)

❑ Tool-body assimilation

- Haruna Kasashima, Wataru Noguchi, Yasumasa Tamura, Masahito Yamamoto, Hiroyuki Iizuka (2024). A Deep Learning Approach to Understanding the Integration of Tools into Body Schema, *ALIFE 2024: Proceedings of the 2024 Artificial Life Conference*

Thank you !

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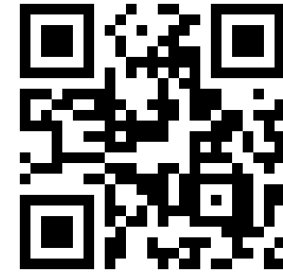
Superposition mechanism as a neural basis for understanding others

[Wataru Noguchi](#) , [Hiroyuki Iizuka](#), [Masahito Yamamoto](#) & [Shigeru Taguchi](#)

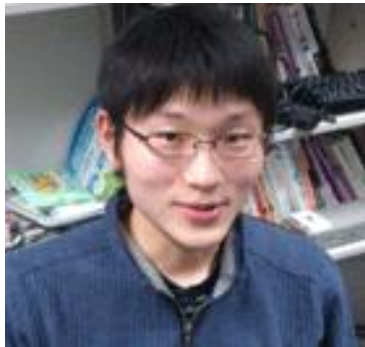
[Scientific Reports](#) **12**, Article number: 2859 (2022) | [Cite this article](#)

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scientific reports



Research movie



Wataru Noguchi

A.I.



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Phenomenology